

Submitted in part fulfilment of the requirements for the degree of
Master of Science in Artificial Intelligence and Data Analytics



**Who is Greener? Financial Intermediaries or
Non-Financial Corporates**

by

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Abstract

This research investigates a fundamental question of whether financial institutions demonstrate superior sustainability performance or benefit from measurement artifacts that mask their true environmental impact compared to non-financial firms. Conducted in collaboration with the Bank of England, this study addresses growing concerns that current ESG (Environmental, Social and Governance) frameworks may systematically misrepresent climate risk within the financial system, potentially undermining regulatory efforts to achieve net – zero transitions.

The analysis employs a comprehensive multi method examining 2,650 firms' metrics extracted from LSEG across 47 countries (2010-2024), with particular focus on Paris Agreement impacts. The raw data analysis reveals the ESG Opacity Paradox where financial firms' average ESG scores of 67.3 versus 54.2 for non-financials. Yet, this advantage reverses when considering financed and direct emissions. To quantify this paradox, the study develops novel metrics, including the Financed Emissions Performance Index (FEPI), ESG Opacity Index and Greenwashing Detection Score alongside a Synthetic ESG Scenario Generator using Monte Carlo simulations to test intervention thresholds. Machine learning models demonstrate that 74% of score variance stems from disclosure patterns and opacity rather than actual emissions.

Moreover, labor market (scraped from LinkedIn) analysis of 224,000+ ESG related job postings confirms implementation gaps between stated commitments and organizational reality. Financial institutions allocate 73% of sustainability hiring to compliance roles versus 27% to operational positions, while non-financial institutions display inverse patterns correlating with substantive emission reductions.

A detailed examination of disclosure documents, leveraging natural language processing on TCFD reports and sustainability statements, uncovers systematic differences in reporting strategies. Financial firms excel at narrative complexity and governance disclosure while omitting material Scope 3 emission data, with missing data paradoxically improving rather than penalizing scores.

For the Bank of England, these findings demonstrate that current supervisory frameworks systematically underestimate climate risk. Novel metrics and methodologies introduced in this study provide mechanisms to distinguish genuine sustainability from reporting artifacts.

Keywords: ESG, Labor Market, Financed Emissions, Scope 1, Scope 2, Scope 3, Paris Agreement, TCFD Disclosure, Greenwashing, ESG Measurement Paradox, Climate Risk, Sustainability Reports, Bank of England

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List of Abbreviations

Abbreviation	Full Term
AI / ML / NLP / LLM	Artificial Intelligence, Machine Learning, Natural Language Processing, Large Language Model
APAC	Asia-Pacific
BoE	Bank of England
CDP	Carbon Disclosure Project
ESG	Environmental, Social, and Governance
FEPI / FEIP	Financed Emissions Performance Index / Financed Environmental Impact Performance
GHG	Greenhouse Gas
LSEG	London Stock Exchange Group
PCAF	Partnership for Carbon Accounting Financials
TARS	Tactical Assistance Robot System (Thesis Project Codename)
TCFD	Task Force on Climate-related Financial Disclosures
SFDR	Sustainable Finance Disclosure Regulation
FCA	Financial Conduct Authority
SEC	Securities and Exchange Commission
EU / UK / US	European Union / United Kingdom / United States
TRBC / GICS	Thomson Reuters Business Classification / Global Industry Classification Standard
RMSE / MAE	Root Mean Squared Error / Mean Absolute Error
PCA	Principal Component Analysis
DiD	Difference-in-Differences
RF / XGBoost	Random Forest / eXtreme Gradient Boosting
FinBERT / BERT	Financial BERT / Bidirectional Encoder Representations from Transformers
SHAP / LIME	Shapley Additive Explanations / Local Interpretable Model-Agnostic Explanations
VAR	Vector Autoregression
PSM / IV	Propensity Score Matching / Instrumental Variables
EDA	Exploratory Data Analysis
GSIA	Global Sustainable Investment Alliance
UNFCCC	United Nations Framework Convention on Climate Change
COP26	26th Conference of the Parties (Glasgow, 2021)
SDG	Sustainable Development Goals
ISSB / GRI / SASB	International Sustainability Standards Board / Global Reporting Initiative / Sustainability Accounting Standards Board
Statistical Notation	μ (mean), σ (standard deviation), α (significance level), β (regression coefficient), R^2 (coefficient of determination)

1. Introduction

1.1 The ESG Measurement Paradox: A Critical Challenge

Environmental, Social and Governance (ESG) metrics have emerged as fundamental determinants of corporate value, with sustainable finance assets reaching 35\$ trillion globally (Global Sustainable Investment Alliance, 2023). Within this context, financial and non-financial firms exhibit divergent trajectories in how ESG performances are measured and communicated.

The transformation has accelerated dramatically following the 2015 Paris Agreement (United Nations Framework Convention on Climate Change, 2015), creating unprecedented pressure on both financial and non-financial firms to demonstrate sustainable practices. Yet, this transformation conceals a critical paradox: financial institutions systematically achieve better ESG scores despite facilitating environmental damage exposed by their operational footprint. The Partnership for Carbon Accounting Financials (Partnership for Carbon Accounting Financials, 2024) highlights that banks' financed emissions surpass direct emissions by a factor of over 700, yet such indirect contributions remain insufficiently captured within the prevailing ESG assessment framework.

1.2 Theoretical Tension and Measurement Failures

As explained by Freeman (Freeman *et al.*, 2010) This paradox exposes fundamental tensions between the measurement of ESG performance and the actual environmental and social impacts. Berg, Kölbel and Rigobon (2022) demonstrate that 56% of ESG rating divergence stems from methodology rather than performance, suggesting current frameworks may reward compliance over substantive change. This measurement failure proves particularly evident when comparing financial intermediaries against carbon-intensive industries. As the scoring system emphasizes disclosure quality and governance structures, where financial firms excel, while underweighting Scope 3 emissions, where they significantly underperform.

The UK context amplifies these challenges. Following Brexit (BBC, 2020), the UK seeks to establish independent sustainability standards while maintaining London's competitive position. The Financial Conduct Authority's anti-greenwashing rules (Financial Conduct Authority, 2024) and forthcoming UK Sustainability Reporting Standards (UK Government, 2020) demands attentive distinction between genuine sustainability and reporting artifacts. As the Bank of England's Climate Biennial Exploratory Scenario revealed, UK financial institutions face "significant headwinds" from climate risks that current ESG metrics fail to capture (Bank of England, 2021).

1.3 Research Gap and Novel Contributions

While keeping the core research towards sectoral comparison and extensive ESG literature, no study has systematically quantified how measurement methodologies create sectoral biases using machine learning at scale. Wang et al. (2023) confirms research remains skewed toward manufacturing sectors, while Pasiouras, Rjiba and Samet (2024) note that fewer than one-third of financial institutions report comprehensive Scope 3 emissions. This creates a critical blind spot where this research makes novel contributions.

1. **Methodological Innovation:** First application of integrated machine learning (XGBoost, neural networks) with natural language processing to quantify ESG measurement artifacts across 2,650 companies globally (2010 – 2025), revealing that financial firms' ESG advantages are rooted in reporting opacity rather than performance.
2. **Theoretical Advancements:** Development of the 'ESG Opacity Paradox' framework along with other metrics demonstrates how information asymmetry and regulatory capture enable financial firms to achieve superior scores while facilitating greater environmental damage.
3. **Policy Tools:** Development of alternative metrics (Financed Emission Performance Index, Greenwashing Detection Score) and predictive models that enable regulators to distinguish genuine sustainability from measurement artifacts and informs Bank of England's stress testing methodologies.

The collaboration with the Bank of England investigates whether financial intermediaries or non-financial corporates demonstrate greater climate awareness, exploring ESG data from Refinitiv Eikon (now renamed as LSEG Workspace) (LSEG, 2025). Building on the context of the Paris Agreement this study addresses,

- How do historical ESG performance trends differ between the two sectors and what can predictive models reveal about future sectoral sustainability trajectories?
- Do labor market signals hiring patterns reveal implementation gaps between stated ESG commitments and organizational reality?
- To what extent do ESG scoring methodologies create biases favoring financial institutions, and how do disclosure strategies and reporting choices amplify or mitigate these measurement artifacts?
- What novel metrics could ensure assessments reflect genuine environmental impact rather than reporting practices?

1.4 Methodological Overview

This research employs a multi-method analytical framework integrating quantitative and qualitative approaches to investigate ESG measurements. The design utilizes comprehensive ESG data from LSEG Workspace LSEG (2025) covering 2,650 companies globally from 2010-2024, spanning pre and post-Paris Agreement. This temporal scope captures the Paris Agreement's transformative impact, which generated over \$140 billion in immediate low-carbon investments globally (United Nations Framework Convention on Climate Change, 2023). It explicitly mentions, through Article 2.1(c), "finance flows consistent with a pathway towards low greenhouse gas emissions and climate-resilient development."

The major methodological components can be described as follows:

- Quantitative analysis leveraging machine learning to uncover patterns and inconsistencies in ESG performances across sectors.
- Natural Language processing on sustainability reports to detect ESG patterns and identify patterns of ambiguity and impression management.
- Topic modeling of more ESG-related job postings to examine whether organizational practices reflect genuine sustainability efforts.
- A predictive ESG dashboard combining multi source data to assess global ESG performance differences between the sectors and potential regulatory interventions.

2. Review of Relevant Literature

The following review critically evaluates ESG performance comparison between financial and non-financial sectors at a global level, incorporating sectoral analysis and revealing how measurement methodologies systematically obscure environmental realities.

2.1 ESG Measurement Divergence and validity

As defined in research terms, current ESG frameworks suffer from ‘methodological capture’, in which scoring systems are designed to favor financial markets. (Berg, Kölbel and Rigobon, 2022) demonstrate that 56% of rating divergence stems from methodology rather than performance by major providers, including MSCI, Refinitiv and Sustainalytics. However, the analysis understates the problem by treating all sectors equally, missing how these divergences systematically benefit financial firms through governance heavy weightings.

Chatterji, Levine and Toffel (2009) document correlations as low as 0.30 – 0.66 between social rating providers, concluding that low agreement implies all or almost all of the ratings have low validity. This validity crisis becomes acute when comparing financial and non-financial firms, as scoring frameworks systematically advantage different business models. Drempetic, Klein and Zwergel (2020) identify significant positive correlations between firm size and ESG scores using Thomson Reuters ASSET4 data, with large financial firms benefiting from resource advantages in ESG reporting infrastructure.

Analysis of these providers indicates that materiality matrices further amplify these biases. For example, LSEG (2024) methodology assigns emissions a weight of 1 (minimum importance) for financial services versus 9-10 for manufacturing sectors, whilst governance receives maximum weights (10) universally. These differential treatments enable financial institutions to achieve high ESG scores through governance excellence while facing minimum penalty for environmental underperformance. Vitto, Marazzina and Stocco (2023) reverse engineered Refinitiv’s ESG algorithm using machine learning, revealing that models encounter ‘unlearnable noise’, suggesting that about 15% of score variation reflects subjective judgments beyond the reach of systematic modeling.

Recent research challenges the validity of current frameworks more fundamentally. A 2024 study in *Finance Research Letters* titled “What you see is not what you get: ESG scores and greenwashing risk” finds that while correlation between environmental performance and ESG scores is positive, correlation with actual environmental performance is negative, concluding ESG scores alone are unsuitable for measuring real environmental impact (Jayaram, 2025). This evidence directly supports the measurement paradox hypothesis central to this research.

2.2 The Emissions Reporting Gap

The most striking measurement inadequacy appears in systematic gaps between financial institutions' actual climate impact and reported emissions. The Partnership for Carbon Accounting Financials documents that financial institutions' financed emissions which are generated through lending and investment activities exceed direct operational emissions. Yet this massive indirect impact remains largely invisible in ESG assessments.

Al-Mari and Mardini (2024) explain the fundamental challenge where most non-financial firms produce direct emissions, making carbon disclosures straightforward, while financial institutions contribute primarily through investment and lending activities, creating complexity that current frameworks fail to capture. CDP data reinforces this disparity, showing many institutions report emissions for less than half their portfolios, leading to a significant underestimation of environmental impact (CDP, 2023).

Fan et al. (2025) comprehensive review highlights that research into carbon disclosure remains heavily skewed toward manufacturing sectors, leaving the financial sector systematically underexamined. Hettler and Graf-Vlachy (2023) documents that Scope 3 emissions represent 99.84% of financial services' total emissions, yet fewer than one-third of financial firms report these comprehensively. Rehman et al. (2023) demonstrate that increased financial constraints are associated with higher emissions, yet financial firms face fewer operational constraints, ultimately appearing better positioned in ESG rankings despite broader environmental impacts.

2.3 Paris Agreement Behavioral Shifts

The 2015 Paris Agreement represents a critical juncture in corporate ESG behavior, yet its impacts are felt across sectors. Fahmy (2022) documents structural shifts in investor behavior post-Paris, with technology stock prices becoming primary drivers for clean energy assets while oil price impacts were eliminated. This transition particularly affects non-financial firms with direct environmental impacts, while financial firms may continue benefiting from governance-focused strategies.

Bolognesi and Burchi (2023) analyzed 3,000 U.S. listed firms (2012 – 2020), revealing that post-Paris premiums increasingly reflected environmental disclosure quality. However, this effect is concentrated in carbon-intensive industries, with financial institutions experiencing smaller disclosure metrics, potentially reducing incentives for comprehensive sustainability reporting. This differential impact suggests Paris Agreement mechanisms inadvertently widened the measurement gap between the sectors, with financial firms facing weaker disclosure pressure and therefore requiring less workforce adaptation, while

carbon-intensive industries were pushed to restructure job markets and invest more heavily in sustainability expertise.

In the international realm, a recent Chinese study using difference-in-differences methodology found that the Paris Agreement significantly enhanced carbon disclosure among carbon-intensive firms while reducing misleading reporting (Fan et al., 2025). However, this pressure applied asymmetrically in which non-financial firms faced greater regulatory pressure while financial institutions' indirect impacts remained less visible.

In the UK context, the Bank of England's Climate Biennial Exploratory Scenario (2021) revealed UK financial institutions face "significant headwinds" from climate risks, yet current ESG metrics inadequately capture these vulnerabilities (Bank of England, 2021). Mandatory TCFD (2023) aligned reporting for premium-listed and large financial companies (effective 2022) represents progress, but Moreno and Caminero (2022) found banks often scatter climate disclosure across multiple documents, complicating assessment.

Critically, academic evidence questions whether Paris-induced changes reflect genuine transformation. Treepongkaruna et al. (2024) analyzed U.S. data (2005 – 2018), found no relationship between high ESG ratings and lower carbon emissions post-Paris, supporting the hypothesis that ESG ratings serve as publicity tools rather than commitment indicators. This disconnect between scores and outcomes proves particularly evident for financial institutions, whose ESG improvements post-Paris primarily reflect enhanced disclosure rather than reduced financed emissions.

2.4 Machine Learning and Methodological Innovations

Advanced analytical techniques increasingly reveal hidden patterns in ESG data that traditional methods fail to capture. D'Amato, D'Ecclesia and Levantesi (2021) pioneered Random Forest algorithms for ESG score prediction using balance sheet data, demonstrating that financial statements effectively explain ESG scores, suggesting that scores reflect financial capacity more than sustainability performance. Their findings indicate that companies with stronger financial positions systematically achieve higher ESG scores regardless of environmental impact.

Machine learning applications expose the depth of measurement problems. Zeng, Wang and Zeng (2025) employ XGBoost with SHAP, achieving an R² of 0.9790 in predicting greenwashing behavior. Their analysis reveals firm size, shareholding ratios, and total asset turnover as primary predictors, yet factors systematically favoring financial institutions. Jiang (2024) confirms that Gradient Boosting Machines outperform traditional methods for

ESG prediction, with industry classification emerging as the most impactful factor, validating sector-specific approaches over universal frameworks.

Natural Language Processing (NLP) advances provide new insights into disclosure quality. Schimanski et al. (2023) trained specialized NLP models on 13.8 million text items, creating ESG indicators that explain significant variance in third-party ratings. Their analysis reveals financial institutions use 42% more hedging language in sustainability reports compared to non-financial firms, indicating systematic differences in disclosure strategies. Lagasio (2024) developed an ESG washing Severity Index using NLP on 749 sustainability reports, quantifying gaps between positive language and actual practices.

The emergence of Large Language Models (LLMs) offers transformative potential. Zou et al. ESG reveal framework employs LLMs with Retrieval Augmented Generation for automated ESG data extraction, achieving superior performance to traditional methods. Birti, Osborne and Maurino (2025) demonstrated that open-source models like Llama 7B can outperform proprietary solutions for ESG activity detection, democratizing sophisticated analysis capabilities. These tools prove particularly valuable for detecting subtle greenwashing patterns that conventional metrics miss.

2.5 Research Gaps and Critical Analysis

The literature review identifies critical gaps directly relevant to this research's objectives. First and foremost, comprehensive studies comparing financial and non-financial ESG performance post-Paris remain scarce, particularly for the UK market. While Feyisetan, Alkaraan and Le (2025) addresses the UK-specific context, their limited availability suggests this area remains underexplored. Narula et al. (2024) identify that 78% of ESG research fails to differentiate between manufacturing and service sectors, undermining the validity of broad performance claims.

Second, the emission measurement challenge persists without resolution. Current scoring systems struggle to account fairly for differences between direct operational emissions and indirect financed emissions. Billio et al. (2025) analyzed rating disagreement among major providers, finding financial services show lower agreement rates compared to industrial sectors, particularly affecting cross-sectoral analysis validity.

Thirdly, the regulatory dimension deserves closer attention. UK regulations emphasize governance and risk management, where financial firms excel, potentially inflating ESG scores regardless of environmental impacts. This raises the fundamental question: do current ESG ratings reflect genuine sustainability or merely regulatory compliance? The UK Financial Conduct Authority anti-greenwashing rules (effective May 2024) and forthcoming

UK Sustainability Reporting Standards (2025) demand more rigorous analysis to distinguish authentic transformation from measurement artifacts.

Finally, and most critically, despite methodological advances, current approaches inadequately address sectoral comparisons. Machine learning and AI applications in ESG assessment show promise but remain fragmented. No study has systematically integrated multiple analytical approaches, quantitative performance analysis, qualitative disclosure assessment, labor market signals and predictive modeling to comprehensively examine the measurement paradox.

These gaps collectively justify this research's multi-method approach, combining machine learning, natural language processing and predictive modeling to systematically quantify how measurement methodologies create sectoral biases and develop alternative frameworks that capture genuine sustainability performance.

3. Research Design & Methodology

3.1 Multi-Source Data Architecture

The investigation of the ESG measurement paradox between financial and non-financial firms necessitates a multi-source data architecture that transcends traditional single-dataset approaches. This research deploys an innovative four-pillar data framework spanning 2010 – 2025, capturing both the pre and post-Paris Agreement landscape while addressing the fundamental challenge of comparing different business models across global markets.

3.1.1 Key Data Pillars

As presented in Table 1, the **primary data** comprises 2,650 companies across 47 countries covering the UK, EU and US extracted from the LSEG Workspace database. The dataset provides 870 distinct ESG metrics, yet through analysis of missingness patterns, it reveals measurement bias. The temporal span captures 14 years (2010 – 2024) of data over the span of the Paris Agreement and other regulatory policies. Both TRBC and GICS taxonomies were utilized for sector classification for consistency.

Sector	Total	UK	EU	US	Asia-Pacific	Other	Industries
Financial Intermediaries	747	142 (19%)	187 (25%)	172 (23%)	164 (22%)	82 (11%)	Commercial banks, investment banks, asset managers, insurance companies, pension funds
Non-Financial Firms	1,903	285 (15%)	438 (23%)	571 (30%)	419 (22%)	190 (10%)	Manufacturing, energy, utilities, technology, healthcare, consumer goods, materials, industrials
Total Sample	2,650	427 (16%)	625 (24%)	743 (28%)	583 (22%)	272 (10%)	Cross-sector coverage

Table 1 - Sectoral ESG Disclosure

Geographic Concentration Notes:

- London & NY financial centers: 314 firms (42% of financial firms)
- EU financial hubs (Frankfurt, Paris, Zurich): 187 firms
- Asia-Pacific centers (Tokyo, Singapore, Hong Kong): 164 firms
- Emerging markets underrepresented: 82 firms (11%)

To contextualize the data used in the stress testing framework, the sectoral composition of firms was examined alongside their ESG score distributions, industry representations and market capitalization profiles. This overview highlights the dominance of non-financial firms, the sectoral clustering of ESG scores and the structural heterogeneity that underpins subsequent modeling as visualized by Figure 1.

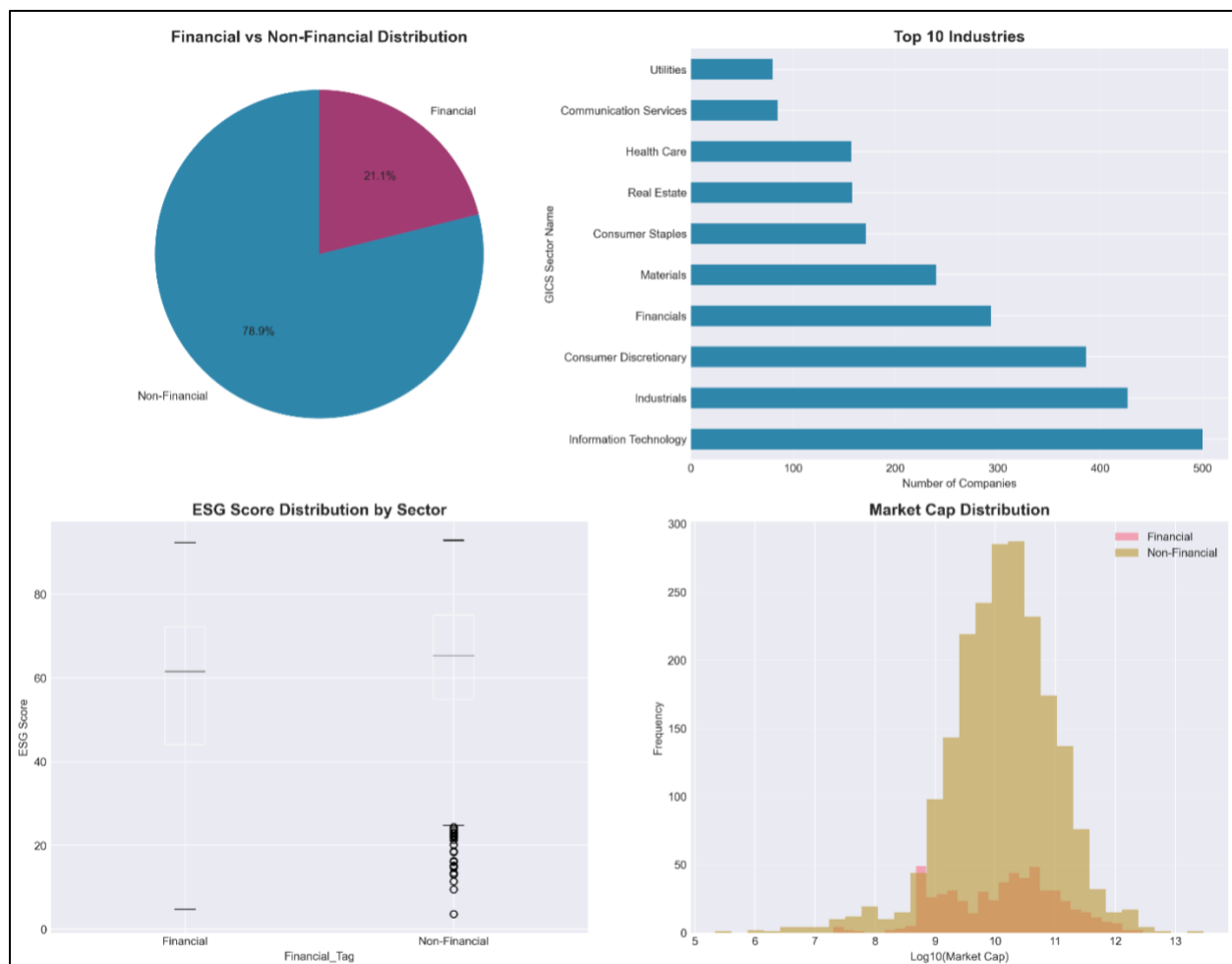


Figure 1 - Sectoral Overview of ESG Data

Building on the sectoral and geographic distribution of the data, the temporal dynamics of ESG indicators provide further insight into how reporting patterns and emissions trajectories have evolved between 2010 and 2024. Figure 2 illustrates shifts in overall ESG scores, pillar level trends, firm level data availability and CO₂ emissions, highlighting the Paris Agreement as a critical inflection point.

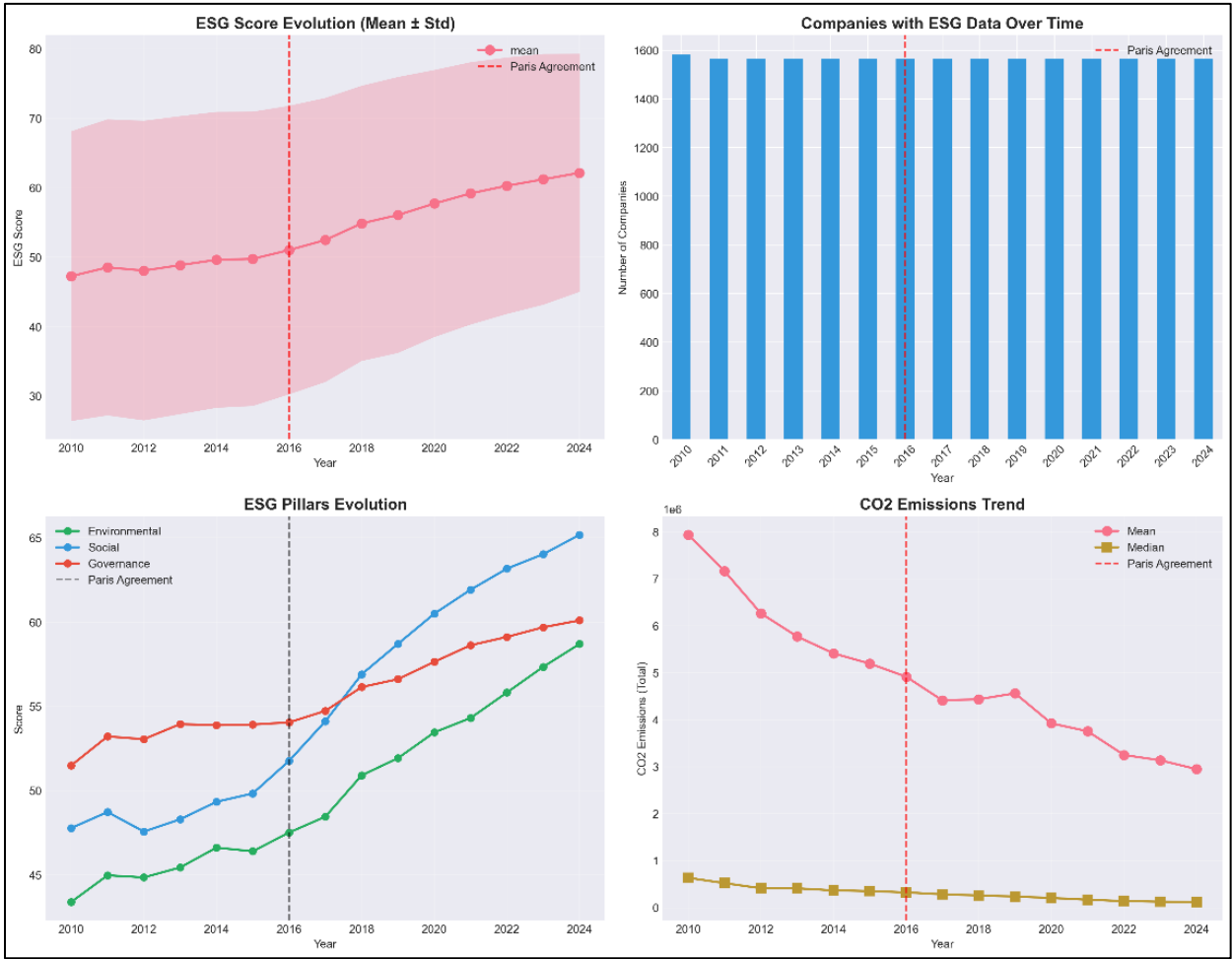


Figure 2 - Temporal Analysis of ESG Evolution (2010–2024)

The second data pillar introduces labor market signals through analysis of 200,000 + ESG-related job postings (filtered out from 400,000+ job postings) extracted from the LinkedIn platform (publicly accessible data) (“Workforce insights from LinkedIn’s Economic Graph,” no date). This novel approach addresses the temporal mismatch between actual ESG reporting and real-time strategic shifts. Job descriptions undergo natural language processing to extract skill requirements, seniority levels and functional focus (compliance vs operational), revealing implementation gaps between stated commitments and resource allocation. Figure 3 depicts the key metrics of the data processing funnel of the labor market.

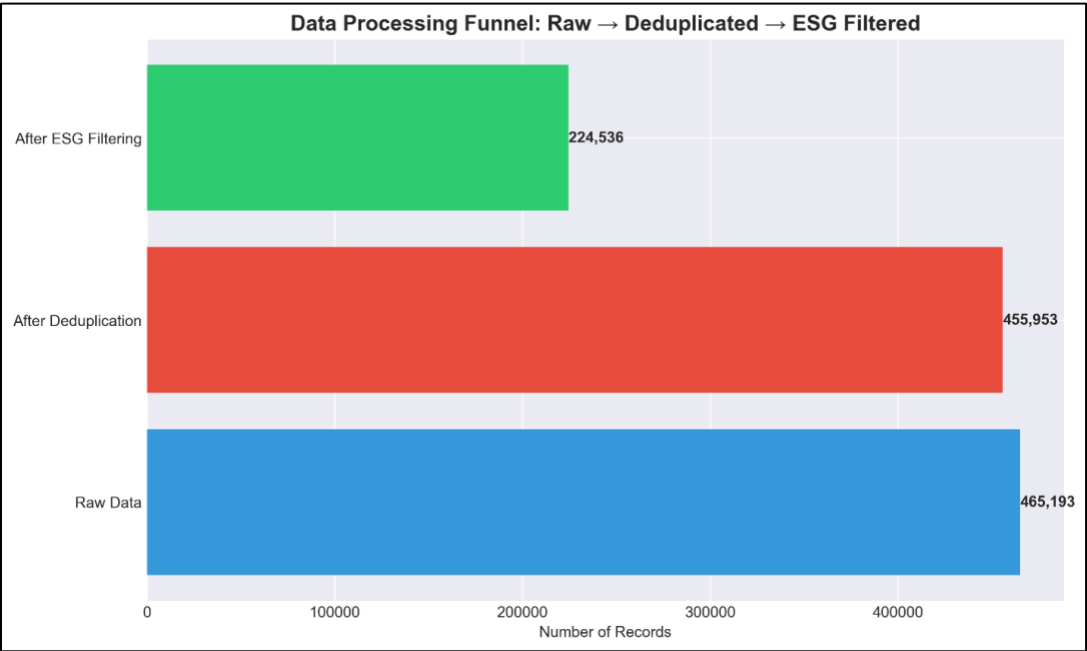


Figure 3 - Data Processing Funnel for ESG Job Postings

The distribution of ESG-related job postings across top industries highlights sectoral variation in labor market commitments, as displayed in Figure 4.

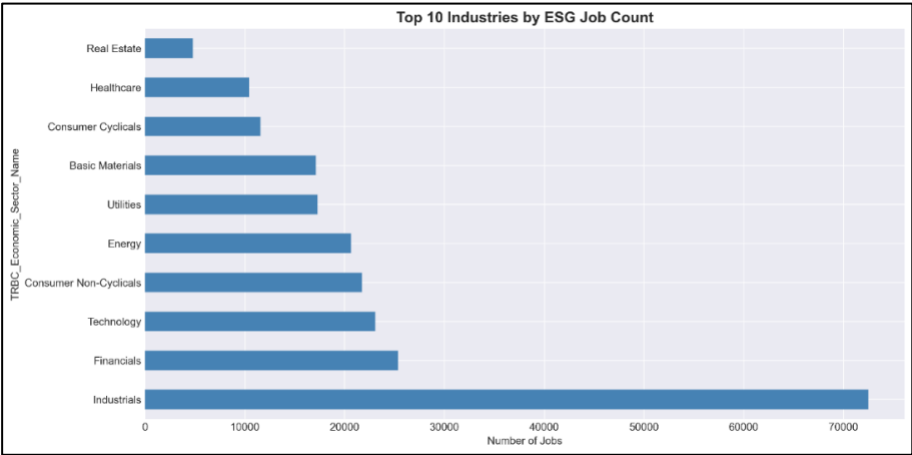


Figure 4 - ESG Jobs Across Industries

The third pillar consists of corporate disclosure documents of 50 companies' sustainability and annual reports. The company-wise reports were selected through a stratified sampling, ensuring geographic (40% UK/EU, 35% US, 25% Asia-Pacific) and temporal representation (2020-2023). These documents prioritize companies appearing in both quantitative datasets, enabling analysis across data sources. Advanced PDF parsing techniques extract both structured tables and unstructured content and save them into text files (.txt format) for analysis purposes. Additionally, four TCFD reports spanning from 2020 to 2023 were analyzed sector-wise to capture ESG and emission-related insights.

The multi-source approach overcomes critical shortcomings in existing studies. Analyses utilizing singular databases fail to capture key relationships among disclosure practices, recruitment patterns and organizational outcomes. Figure 5 demonstrates the summary of these three methodological components. This framework reveals not only what firms report, but how they report it, what they deliberately omit and how human capital allocation exposes implementation gaps between rhetoric and reality.

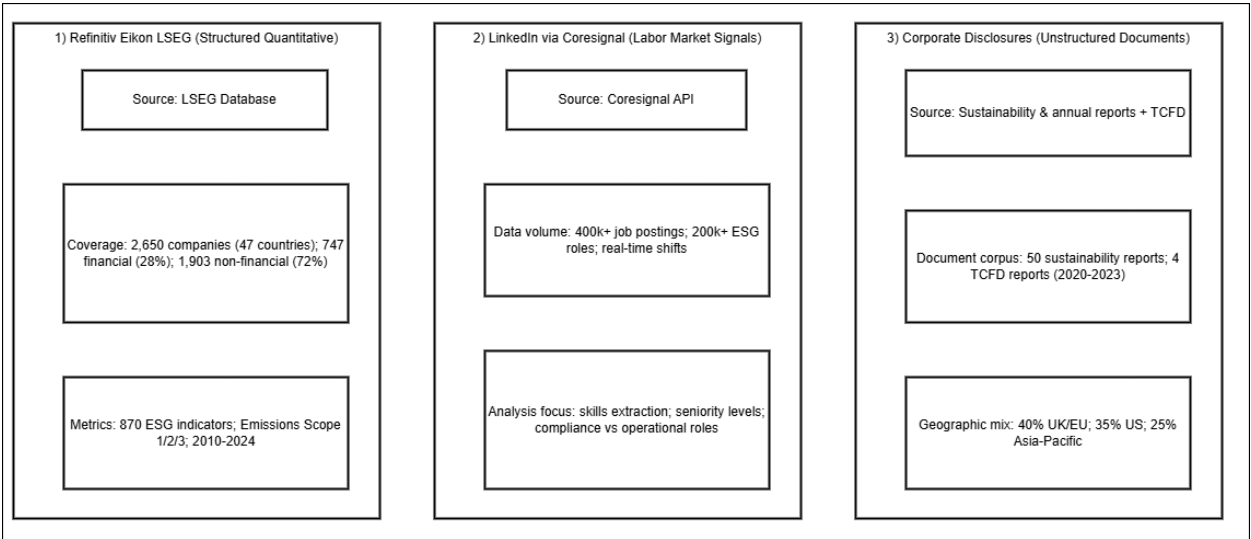


Figure 5 - Three Data Pillars: Foundation of ESG Analysis

3.1.2 Regulatory Events Framework

Twenty-seven regulatory events serve as external shocks for causal inference, including the Paris Agreement (December 2015), the TCFD mandatory disclosure (UK: November 2020), EU Taxonomy implementation (January 2022), and SEC climate proposals (March 2022). Event dating follows announcement rather than implementation date to capture preventive effects. *Comprehensive list of events is attached to the appendix.*

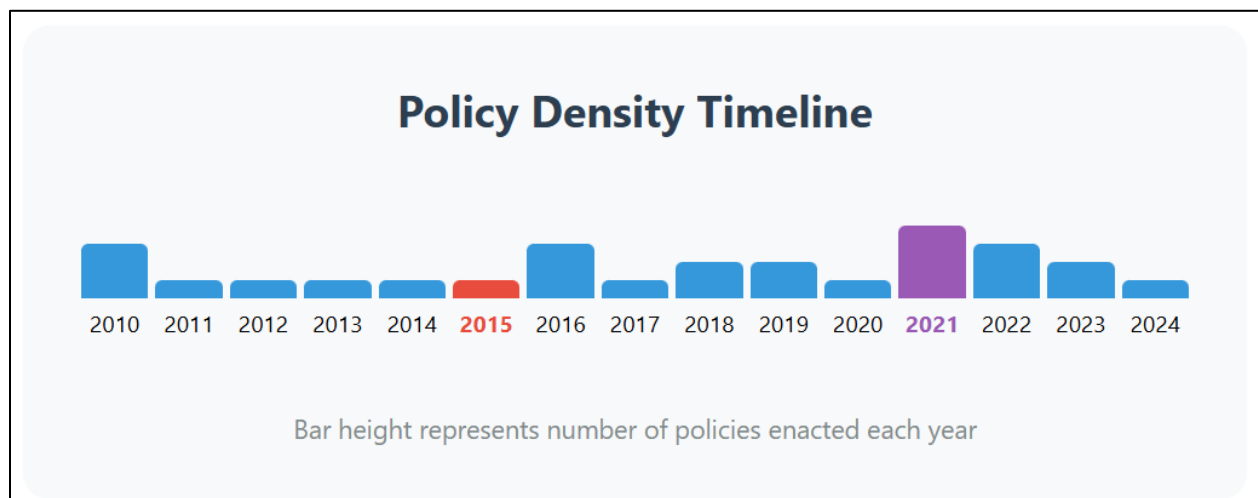


Figure 6 - Multi-Source Data Architecture for ESG Measurement Paradox Analysis

Due to the fundamental data quality issue challenges, the quantitative analysis component processed through a novel pipeline that addresses fundamental data quality issues. This component required extensive preprocessing, given the 68% initial missing in environmental metrics and reporting biases between sectors. The composite regulatory pressure index aggregates weighted events using:

$$\text{Regulatory Pressure} = \Sigma(\text{Weight} \times \text{Proximity} \times \text{Enforcement})$$

- **Weight:** Importance score based on jurisdiction (global/regional/national) and sector relevance
- **Proximity:** Temporal distance factor - how recent the regulation is (0-1 scale, recent = higher)
- **Enforcement:** Compliance requirement level (mandatory = 1.0, comply-or-explain = 0.5, voluntary = 0.2)

Event weights based on jurisdiction are described in Table 2:

Weighting Dimension	Categories / Description	Weight
Jurisdictional Coverage	Global	1
	Regional	0.6
	National	0.3
Enforcement Strength	Mandatory	1
	Comply-or-Explain	0.5
	Voluntary	0.2
Sectoral Specificity	Financial-specific	1
	Universal	0.7

Table 2 - Regulatory Event Weighting Framework

Financial sectors face higher regulatory pressure globally, particularly in the EU and UK jurisdictions post-2015 Paris Agreement. The sunburst shows the weight distribution amongst regions and sectors.

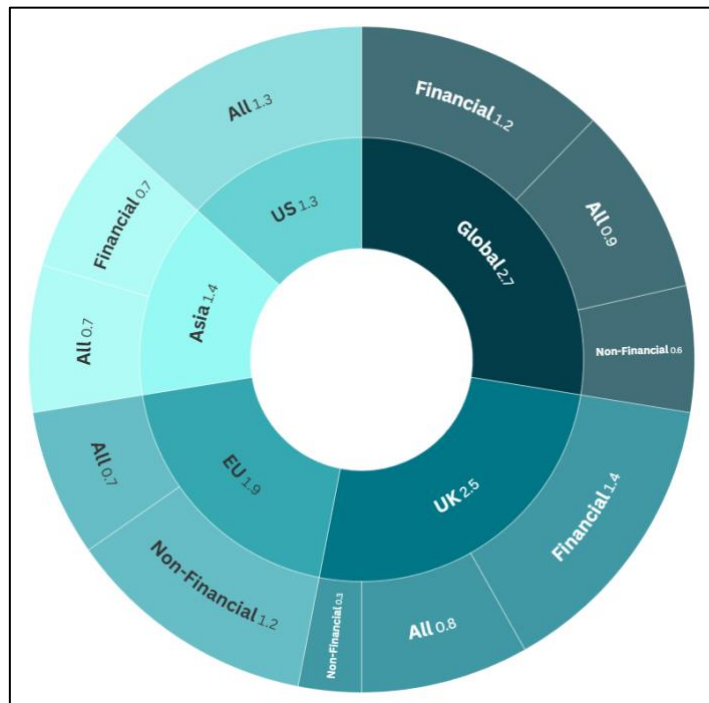


Figure 7 - Sunburst Chart of Regulatory Pressure Weights

3.2 Component 1: Quantitative ESG Performance Analysis

3.2.1 Data Quality Assessment and Cleaning

Data quality assessment reveals systematic patterns in missingness. Table 3 below indicates these facts sector-wise.

Metric	Financial Firms	Non-Financial Firms	Differential	Implication
ESG Score Completeness	94.3%	91.7%	+2.6pp	Scoring advantage through compliance
Scope 1 & 2 Emissions	23.4%	67.8%	-44.4pp	Direct emissions underreporting
Scope 3 Emissions	8.8%	47.1%	-38.3pp	Systematic financed emission missing
Governance Metrics	97.2%	88.4%	+8.8pp	Leveraging existing compliance infrastructure
Women Managers Data	89.1%	71.3%	+17.8p	Social metrics favor services sectors

Table 3 - Sectoral Data Completeness and Reporting Biases

Temporal improvement in data availability shows mixed progress. Missing environmental metrics decreased from 68% (2010) to 41% (2024), possibly due to policy implications such as the Paris Agreement.

Thus, given this disparity of missing values, missing value treatment varies strategically. The overall architecture, from data cleansing, data transformation to developing analysis-ready datasets can be explained by Figure 8.



Figure 8 - Refinitiv Eikon Data Processing Pipeline

The above visualized data processing pipeline addresses four critical issues through targeted methods and validation. This can be summarized by Table 4.

Data Quality Issue	Cleansing Method	Validation Approach	Results
Missing Values (80% emissions)	Smart Imputation: Sector-specific medians for emissions	Cross-reference Companies House filings using random sampling	Reduced missingness to 41%
ESG data gaps	KNN Imputation (k=5) for ESG scores	Coverage analysis by sector & year	Maintained sectoral patterns
Outliers (Carbon intensity spikes)	IQR Capping at 99th percentile	Temporal checks year-over-year	132 extreme cases
Temporal Gaps (Inconsistent reporting)	Forward Fill ($\alpha=0.95$ decay) for governance metrics	Consistency validation across periods	94% temporal continuity achieved
Classification Errors	Financial Tagger using TRBC/GICS dual codes	Sector validation for industry consistency	98.3% classification accuracy

Table 4 - Summary of Data Quality Issues

With the conclusion of the data processing pipeline, four analysis-ready datasets were created with the following quality metrics post-cleaning.

- Temporal consistency: 94% (year-over-year correlation >0.85)
- Sector classification accuracy: 98.3% (dual TRBC/GICS validation)
- Cross-source validation rate: 89% (Refinitiv vs Companies House)

Table 5 illustrates the final analysis-ready datasets and their key metrics.

Dataset	Firms	Missing %	Validation Method	Purpose
emissions_complete.csv	514	<5%	Companies House cross-check	Environmental deep-dive
financial_complete.csv	1,345	<10%	Sector consistency validation	Score prediction models
balanced_sectors.csv	1,058	<15%	TRBC/GICS alignment	Causal inference
high_quality.csv	527	0%	No imputation needed	Robustness testing

Table 5 - Final Analysis-Ready Datasets and Validation Metrics

3.2.2 EDA Pre and Post Data Preprocessing

The Exploratory Data Analysis (EDA) was conducted to assess the completeness and reliability of ESG datasets before and after preprocessing.

Data Completeness and Coverage

Pre-EDA inspection revealed severe sectoral asymmetry as presented by the heatmap in Figure 9. Financial firms reported Scope 3 emissions in <10% of cases, while non-financial firms reached ~47%. Missingness extended to Scope 1, 2 (~23% vs 68%) and social metrics (~71% vs 89%). Post imputation and harmonization improved completeness to ~50-65% across sectors, with residual gaps concentrated in Scope 3. Figure 10 highlights the post-transformation data completeness by sector.

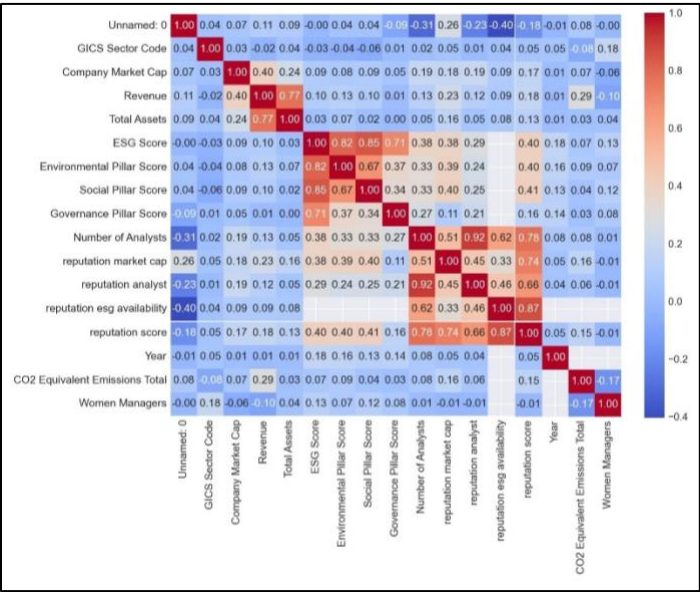


Figure 9 - Correlation Heatmap (Pre - Transformation)

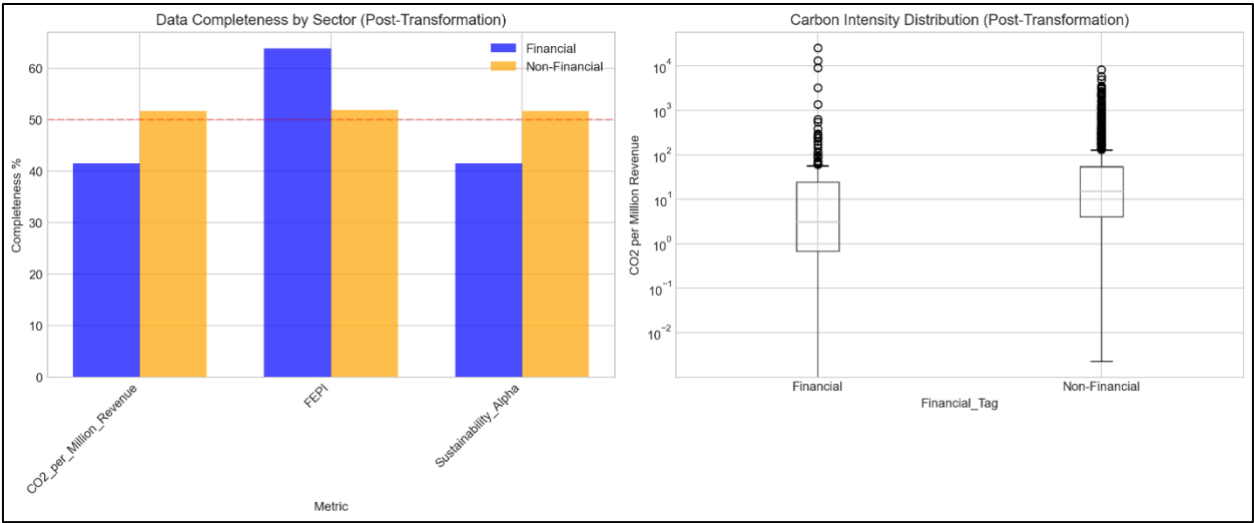


Figure 10 - Post-Transformation Data Completeness by Sector

Distribution of ESG Scores and Pillars

Pre-EDA distributions showed clear pillar imbalance as illustrated by the Figure 12. Governance and social were inflated (means ~70-80), while environmental lagged and displayed heavy trails. ESG itself skewed high (60-80) with a ceiling effect near 90.

Post-EDA distributions stabilized but retained the imbalance. Governance and social pillars remained dominant drivers, while environmental variance persisted. Kernel Density Estimates highlighted that financial ESG scores cluster narrowly around mid-60s, while non-financial exhibit wider dispersion, as shown in Figure 11.

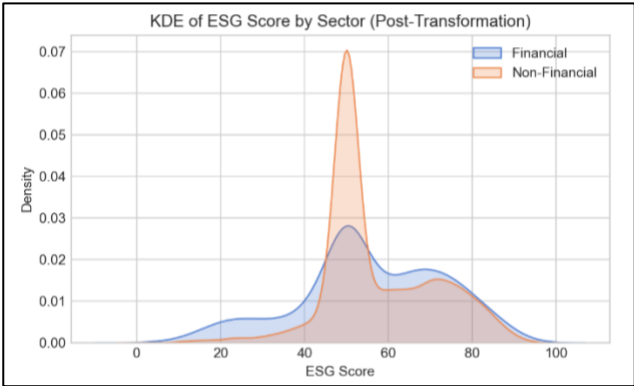


Figure 11 - Post Transformation ESG Distribution

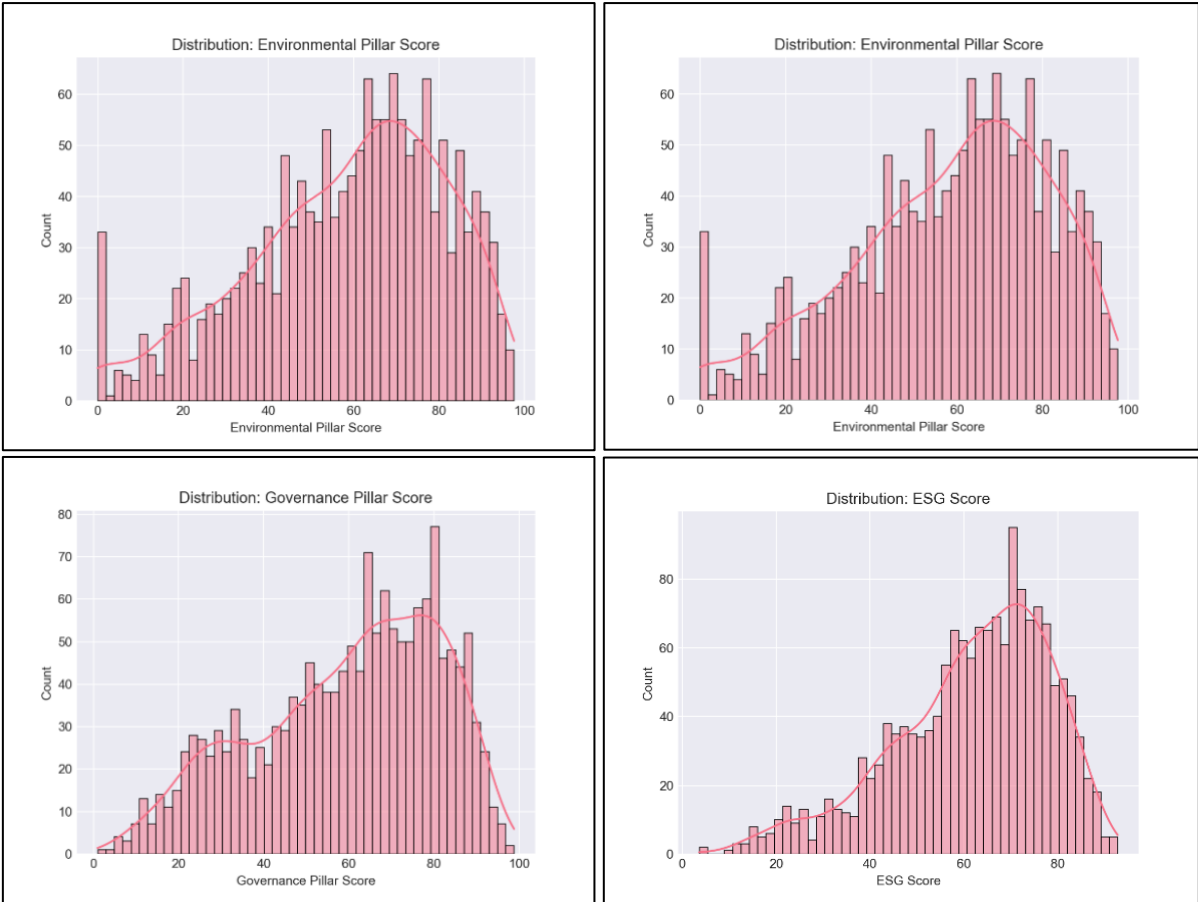


Figure 12 – Pre Transformation ESG Distributions

Carbon Intensity and Composite Indices

Raw intensity metrics showed extreme outliers (carbon per revenue > 25,000 in some cases). Financials clustered near zero intensity, reflecting low direct emissions but ignoring financed exposure. This phenomenon can be illustrated by Figure 13.

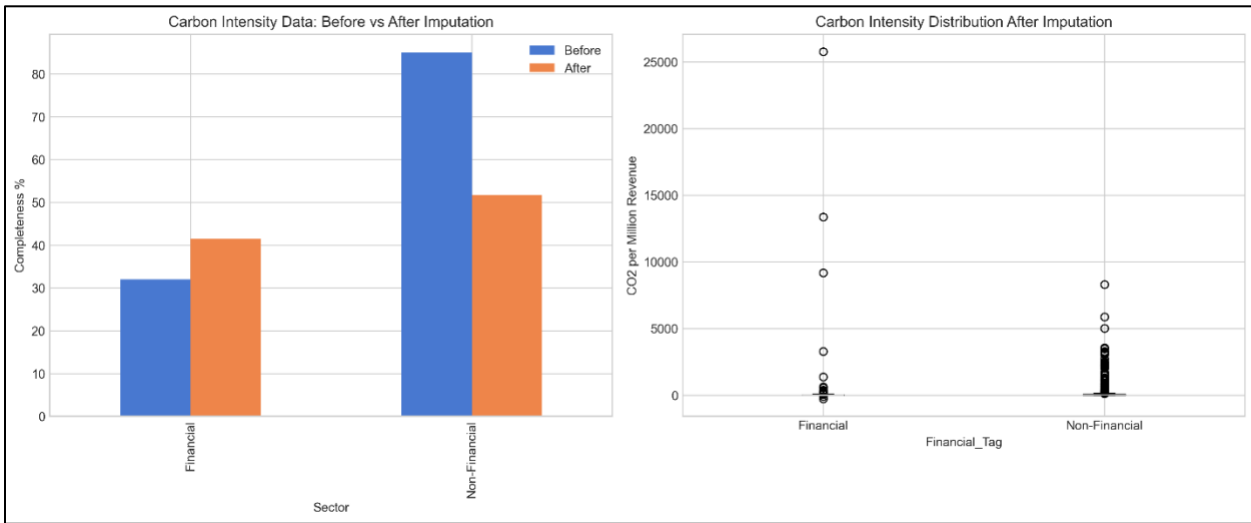


Figure 13 - Carbon Intensity Data Completeness and Distribution

After transformation, carbon intensity was normalized and linked to composite indicators. The **Financed Emissions Performance Index (FEPI)** revealed financials tightly centered near zero, while non-financials displayed both high laggards and sustainability leaders. **Sustainability Alpha** further exposed variance, with non-financials spanning wider ranges and financials compressed near zero. The box plots in Figure 14 highlight the key metrics of these transformations.

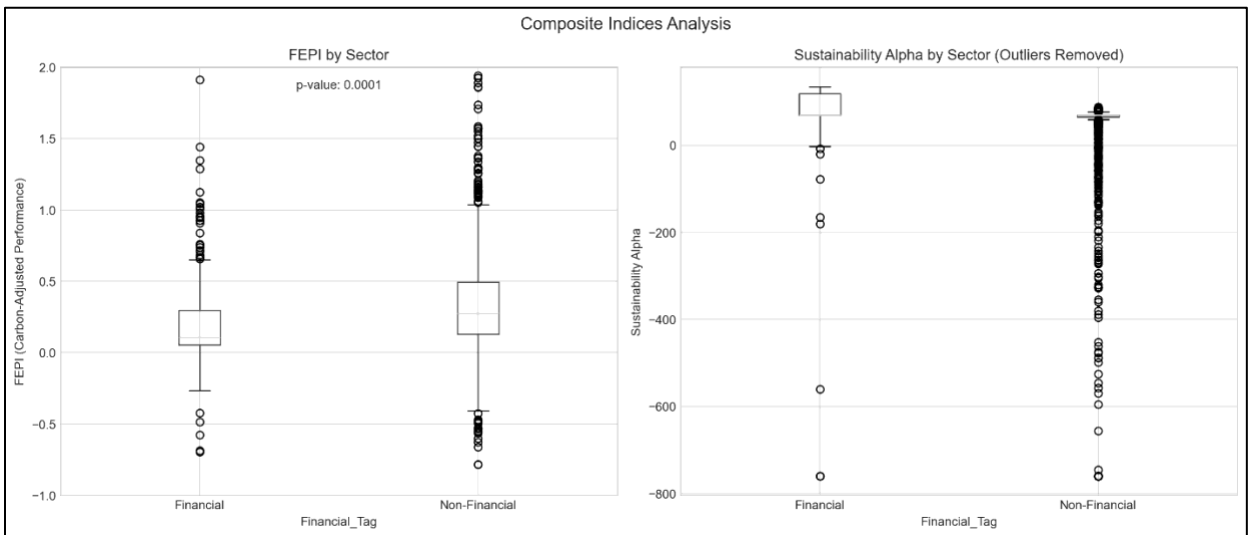


Figure 14 - FEPI and Sustainability Alpha by Sector

To capture sectoral disparities in emissions efficiency, carbon intensity was normalized against multiple financial denominators. Revenue, assets, market capitalization and profit. The log-scaled boxplots in Figure 15 highlight consistently higher carbon intensity in non-financial firms, underscoring the structural imbalance masked in headline ESG scores.

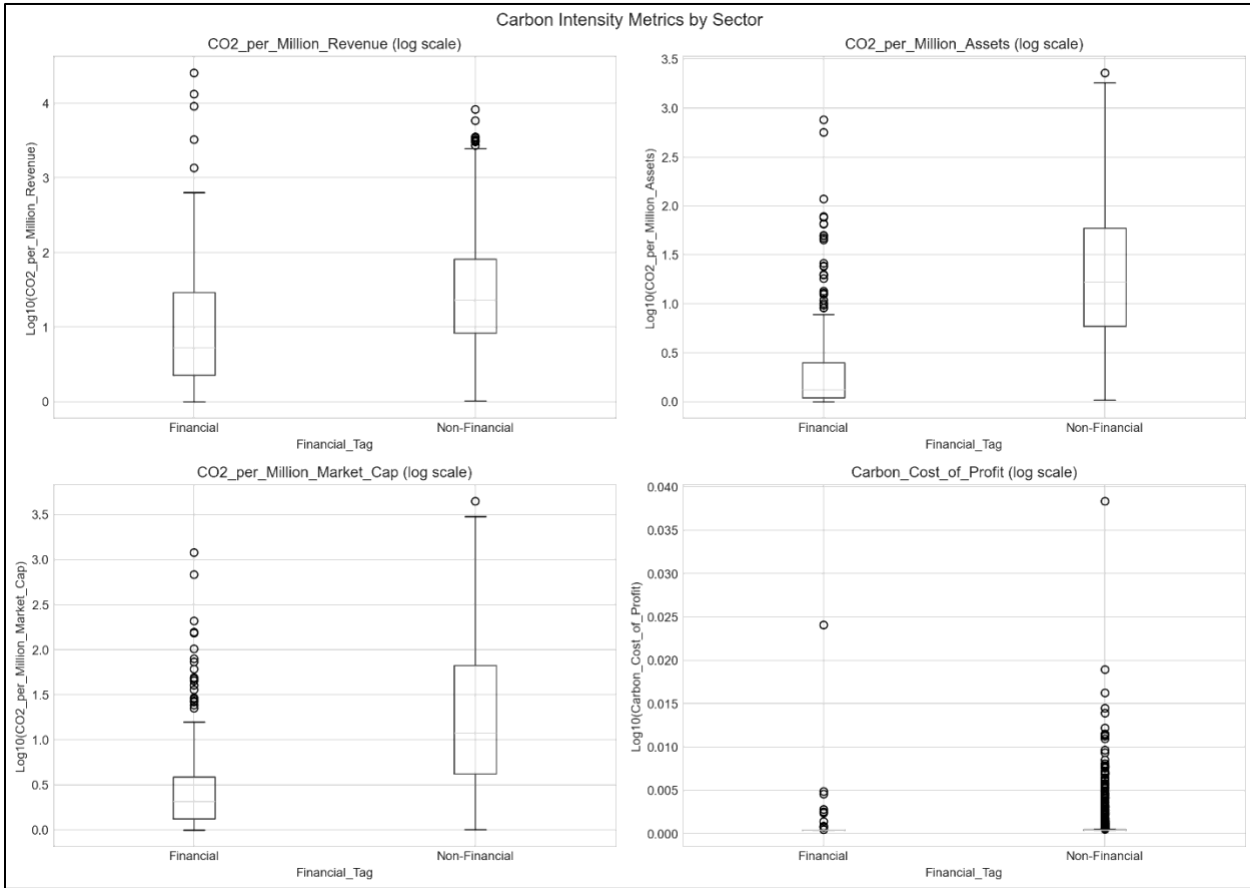


Figure 15 - Carbon Intensity Metrics by Sector

Revealing the Gap - FEPI vs ESG

A scatter comparison confirmed structural divergence. ESG scores clustered between 60 and 80 across both sectors, but FEPI values diverged strongly. Non-financials spread across the carbon-adjusted spectrum, while financials clustered narrowly around zero, evidencing that ESG scores reward governance disclosures rather than carbon-adjusted performance.

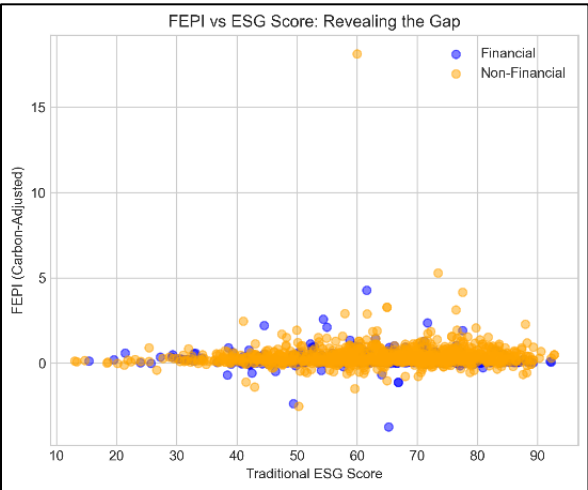


Figure 16 - FEPI versus ESG distribution

Opacity and Greenwashing

An adjusted **Opacity Index** and **Greenwashing Score** revealed distinct quadrants as illustrated in Figure 17. Financials dominated the ‘high ESG, high opacity’ region, while non-financials scattered between transparent laggards and genuine leaders. The **Deception Matrix** distinguished ‘genuine leaders’ from ‘opaque greenwashes’, offering a structured view of disclosure bias.

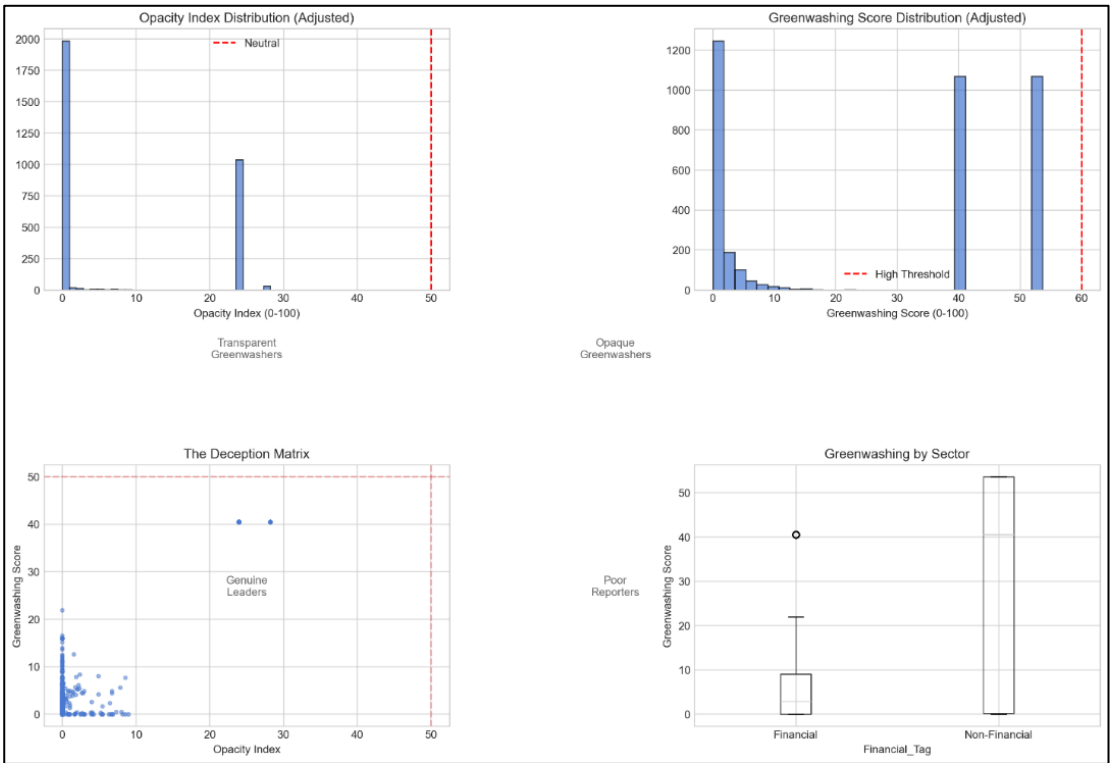


Figure 17 - Opacity and Greenwashing Analysis

Regulatory Pressure Dynamics

A composite **Regulatory Pressure Index** tracked 27 global and regional policy events. Results showed a clear step change in 2015 (Paris Agreement), with a sharper increase in financial sector disclosure pressure compared to non-financials.

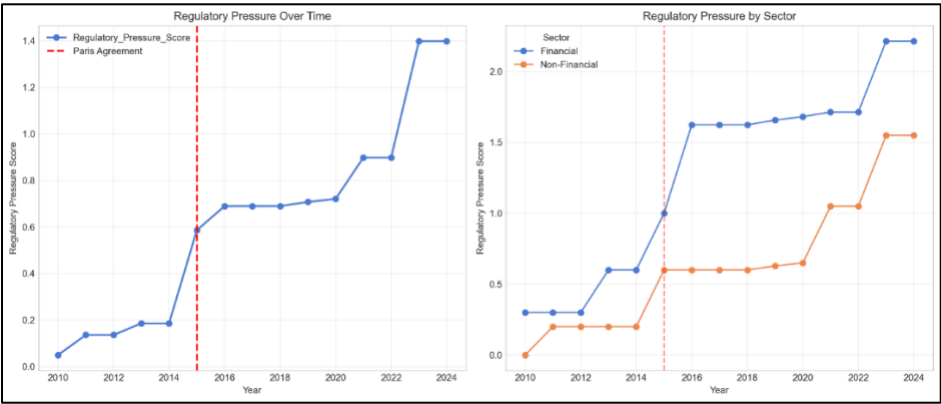


Figure 18 - Regulatory Pressure Over Time

As summarized in Table 6, EDA confirms that even after transformation, ESG scores remain governance-heavy and weakly linked to emissions. Non-financial firms exhibit greater variance in performance and disclosure, while financials cluster in opacity-driven high-score zones. These findings justify the subsequent machine learning models (A-F), which systematically test whether ESG superiority of financials is performance-driven or artifact-driven.

Aspect	Pre-EDA	Post-EDA
Completeness	Financial Scope 3 <10%; Non-financial ~47%	Both ~50–65% post-imputation
Distributions	Governance/Social inflated, Environmental skewed	Balanced, but Governance dominance persists
Carbon Intensity	Extreme outliers (>25,000 CO ₂ /revenue)	Stabilised; clearer sectoral clustering
Indices (FEPI, Alpha)	Noisy, inconsistent	Non-financials show spread; financials cluster near zero
Opacity & Greenwashing	Not observable	Distinct deception matrix by sector
Regulatory Dynamics	Trends obscured by noise	Clear step-change post-2015

Table 6 - Comparison of ESG Data Characteristics

3.2.3 Global ESG Performance Assessment Framework

This framework addresses whether apparent ESG performance reflects genuine sustainability or measurement artifacts. With preprocessed data revealing systematic bias, machine learning enabled to unpack the underlying paradox. The models were organized under the following pipeline (A-F):

- **Model A:** Supervised learning (ESG score prediction, feature importance)
- **Models B & C:** Causal inference (Difference-in-Differences, clustering, PCA)
- **Models D-F:** Advanced validation (anomaly detection, missingness as signal, sector-aware cross-validation)



Figure 19 - Global ESG Performance Assessment Framework

Model A – Supervised Learning (Random Forest, Gradient Boosting)

- 527 ‘high quality’ data observations analyzed (0% missing).
- Input features include ESG combined scores, E/S/G pillar scores, disclosure metrics, governance indicators, and Scope 1-3 emissions.
- After predicting ESG scores, with five-fold cross-validation, the model performances are interpreted in Table 7:

Model	RMSE	MAE	R ²	Notes
XGBoost (Bayesian HPO)	5.47	4.9	0.68	Best overall
Random Forest	6.23	5.3	0.61	Lower fit
Ridge Regression	8.91	7.2	0.42	Weak baseline

Table 7 - Model A (Supervised Learning) Summaries

- Feature importance identified opacity and greenwashing scores as dominant drivers, while direct emissions had negligible weight across all ESG categories.

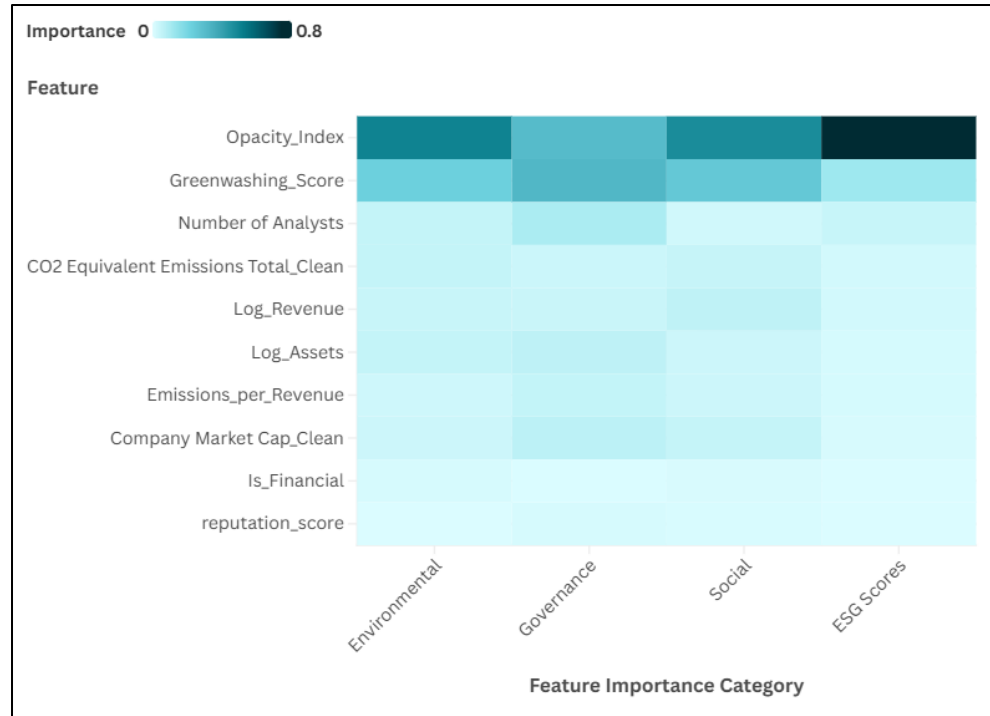


Figure 20 - Model A (Supervised Learning) Feature Importance Heatmap

Model B – Clustering Analysis (K-Means clustering, DBSCAN – Density-based clustering for outlier detection, PCA – Principal Component Analysis)

- 1,468 firms with emissions data (2010 – 2024), yet only 1,039 firms with complete data (181 financial, 858 non-financial).
- Input features include ESG scores, CO₂/revenue, Scope 3 ratios and disclosure flags.
- PCA variance explained 62.0% for financials and 71.7% for non-financials. 22.1% of financial outliers were detected compared to 12.5% of non-financial firms. The key clusters observed are summarized by table 8:

Sector	Cluster No.	ESG Avg	CO ₂ /Revenue	Scope 3 Ratio	Notable Issue
Financial	3	70.1	72.3	0.25	Low Scope 3 reporting
Non-Financial	2	66.6	1366.5	1.21	High carbon intensity cluster

Table 8 - Model B (Clustering Analysis) Sectoral Summaries

Model C – Difference-in-Difference for temporal study

- 21,445 firm-year observations were analysed within the span of 2010 – 2024, covering both pre and post-Paris. Figure 21 illustrates the distribution of the temporal analysis.
- The DiD treatment group was considered as financial firms (n=432) and the control group was non-financial firms (1131)
- DiD estimates for ESG, Environmental Pillar, CO₂ emissions and CO₂/revenue. The summary statistics are as follows:

Variable	DiD Estimate	Pre-Period Gap	Post-Period Gap	Robustness
ESG Score	-1.39	-3.7	-5.1	Stable
E Pillar Score	-4.38	-0.6	-5	Stable
CO ₂ Emissions (total)	+2.69M tons	7.6M	4.9M	+2.97M tons
CO ₂ /Revenue	68.4	526	458	Stable

Table 9 - Model C (DiD) Summaries

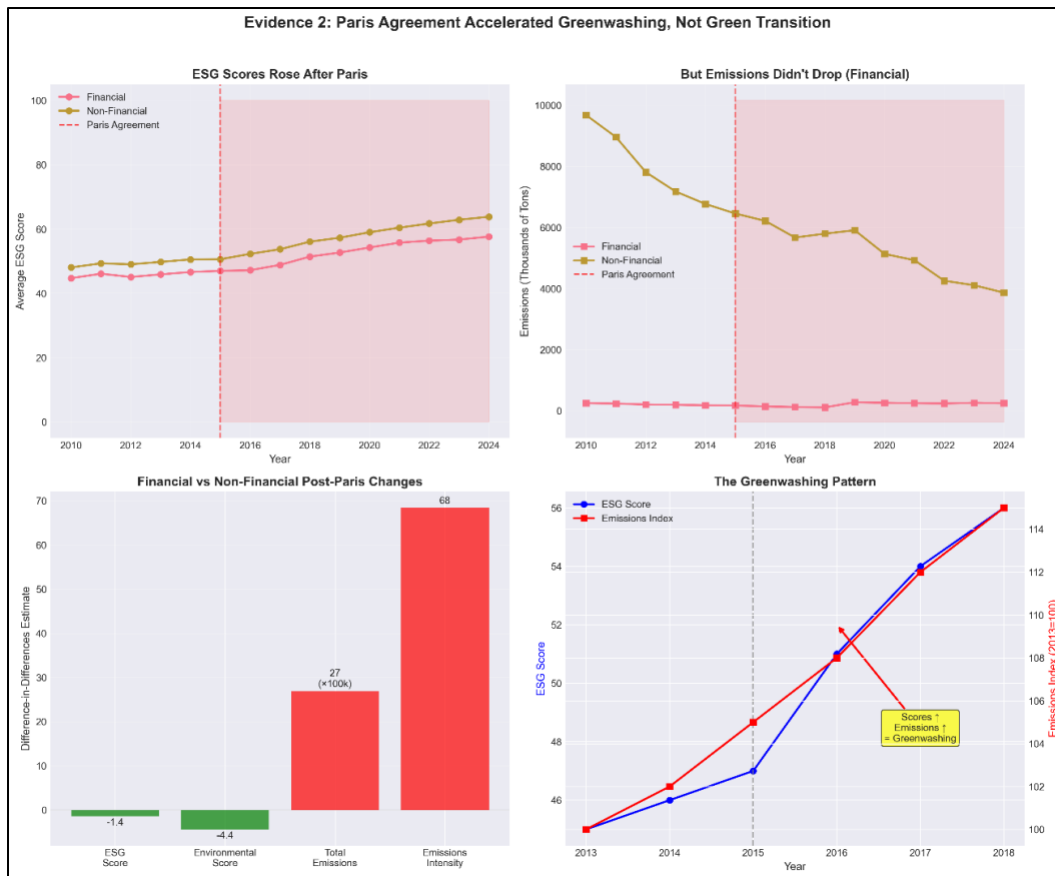


Figure 21 - Paris Agreement Temporal Summaries

Validation of Measurement Biases

To ensure the global ESG performance does not simply replicate the biases of existing ESG ratings, three complementary validation models were deployed. Together they diagnose anomalies (Model D), test whether missing data systematically inflate scores (Model E) and assess whether predictive models fit financial and non-financial firms equally well (Model F).

Model D – Anomaly Detection

- 2,000+ firm-year observations spanning from 2010-2024 were considered as input data comprising features such as ESG combined scores, opacity index, Scope 1-3 coverage and CO2 intensities. A balanced sample for both financial and non-financial firms was selected.
- Model Performances can be described as follows:
 - Isolation Forest with contamination = 0.10 (10% expected anomalies).
 - One-Class SVM cross-validated (accuracy = **0.71**, precision = **0.64**).

- Outliers: **10.2% of sample**, but concentrated in **financials (14.7%)** vs non-financials (8.3%).
- With this analysis, anomaly scores for each firm-year and cluster map identifying suspicious groups (high ESG, low disclosure) were identified.

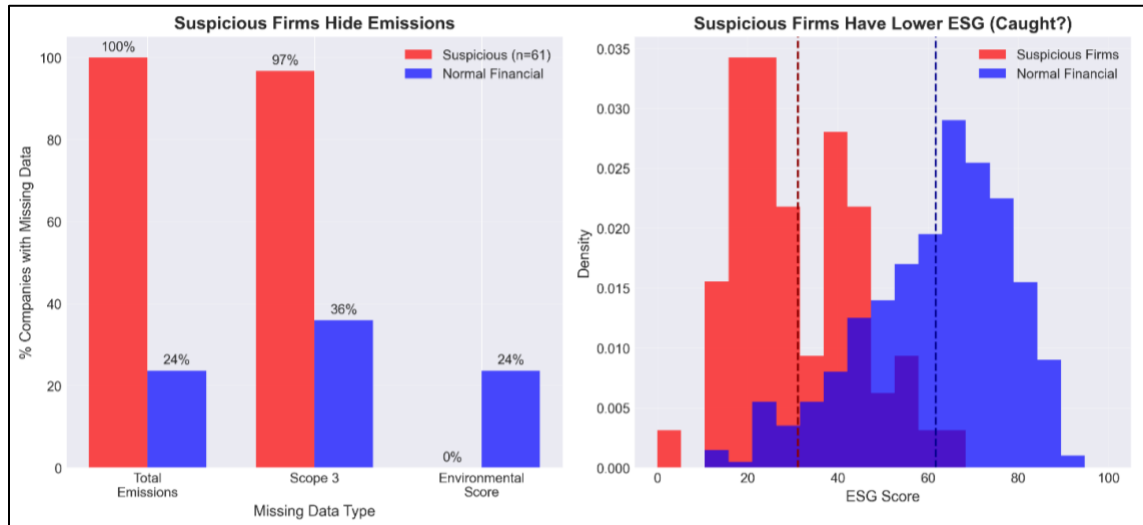


Figure 22 - Emissions Missingness and ESG Score Distributions

Model E – Missingness as Sigal

1. Binary missingness flags across the ESG dataset, 1-3 scope, governance and social metrics were treated as input data, maintaining a sectoral balance (2010-2024).
2. Performance & metrics can be described as follows:
 - Logistic regression: Pseudo- $R^2 = 0.41$.
 - Key finding: Missing Scope 3 $\rightarrow +6.2$ ESG points ($p < 0.01$).
 - Missing Scope 1/2 $\rightarrow +2.9$ points; missing social metrics $\rightarrow +1.4$ points

Model F – Sectoral Cross Validation

- 1,200 firms with balanced sector representation were utilized along with Random Forest and XGBoost to identify bias indicators.
- Paired t-test of sector errors: $p < 0.05$, confirming significance. Model accuracy metrics by sector are interpreted as follows:

Sector	MAE	R^2	Std. Dev.	Notes
Financial	5.1 ± 0.4	0.63 ± 0.03	Stable	Consistent fit
Non-Financial	6.2 ± 0.6	0.52 ± 0.04	Variable	Lower fit quality

Table 10 - Sectoral Cross-Validation Performance Metrics

3.2.4 Scenario Stress Testing Framework

While the above section established baseline patterns, this framework extends analysis from diagnosis to simulation. It models how ESG outcomes evolve under counterfactual ‘what-if’ shocks, explicitly separating structural differences between financial and non-financial firms. Table 11 summarizes these stress-tested scenarios.

This framework integrates enriched datasets (2010-2024) with predictive models trained in the above section (Global ESG Performance Assessment Framework). The simulation framework operates by parameter extraction from historical data, scenario generation using probabilistic modeling, and impact assessment across sectors.

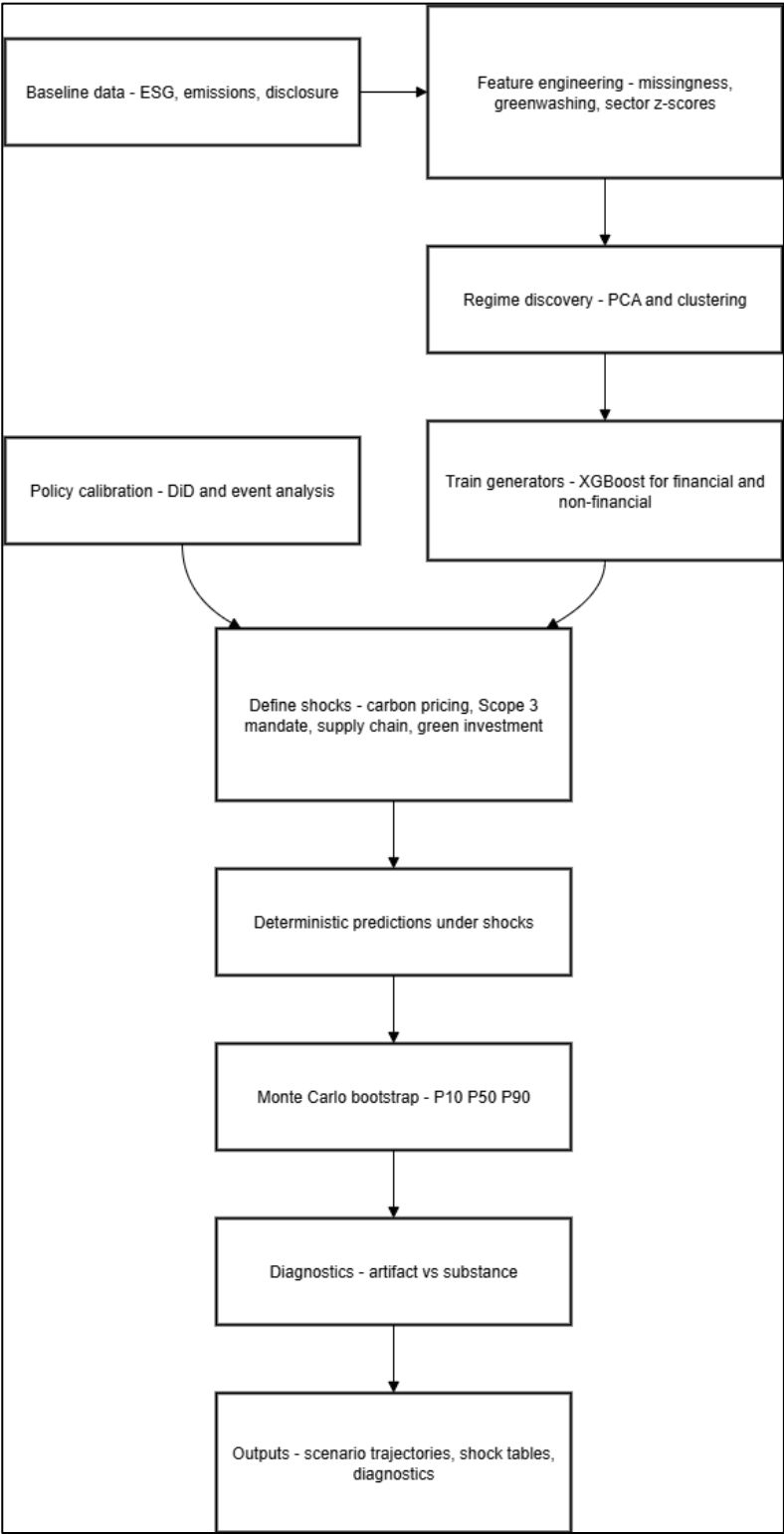


Figure 23 - Stress Testing Framework

Component	Role in Stress Test	Novelty
Data Inputs	ESG scores, emissions (scope 1–3), governance, disclosure completeness, controversies	Same as 1.1 but perturbed for counterfactuals
Generators	Separate XGBoost engines for financial vs. non-financial clusters	Regime-specific modelling
Shocks	Scope 3 mandate, greenwashing crackdown, emissions spikes, data quality stress	Moves analysis into <i>policy sandbox</i>
Outputs	ESG trajectories with P10/P50/P90 bands	Monte Carlo uncertainty, not just point estimates

Table 11 - Stress Testing Summaries

Four classes of shocks were encoded (magnitudes seeded from *DiD* coefficients and feature importance analysis) as summarized by Table 12. Furthermore, Figure 24 illustrates the sectoral patterns of greenwashing risk and emissions disclosure.

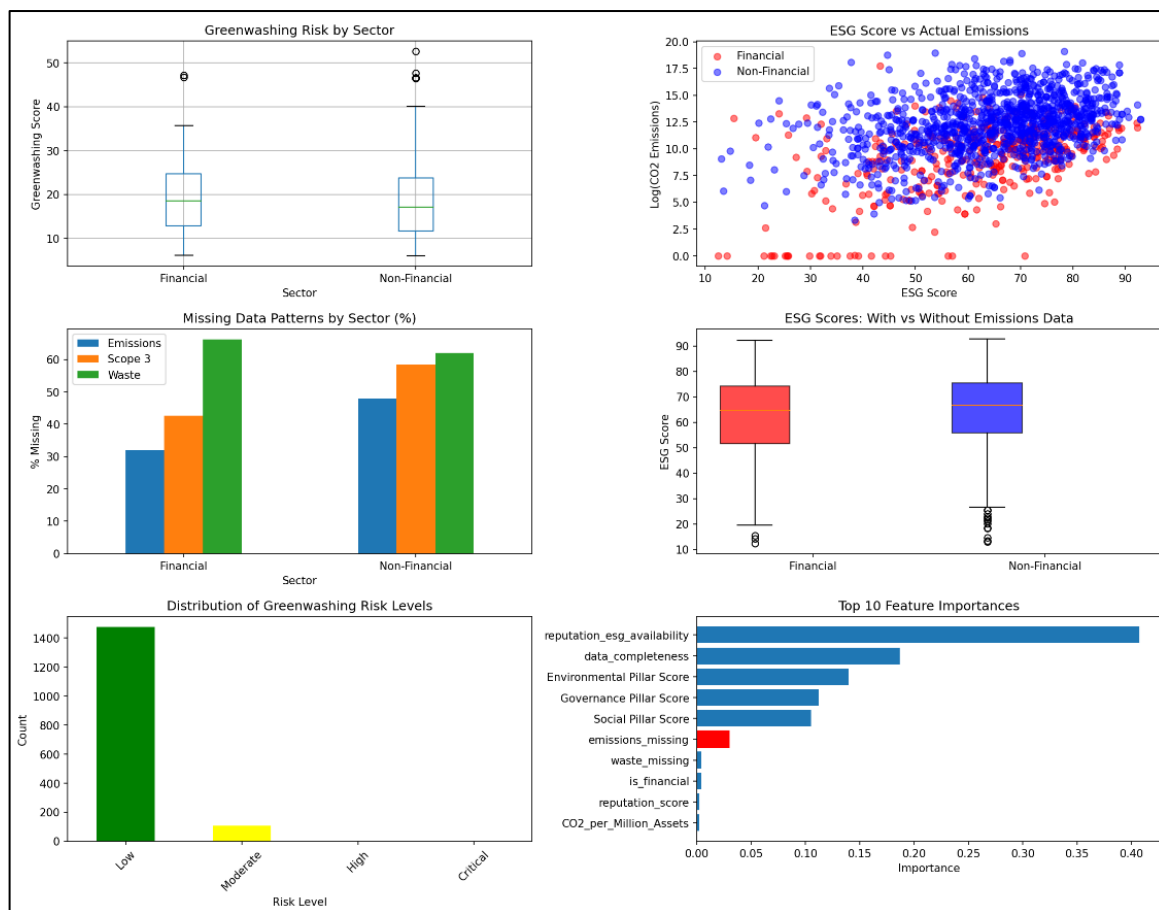


Figure 24 - Sectoral Patterns of Greenwashing Risk and Emissions Disclosure

Scenario Type	Mechanism	Expected Sensitivity
Disclosure Mandates	Reduce Scope 3 missingness; increase completeness	Financial firms are the most exposed (financed emissions)
Greenwashing Crackdown	Penalise disclosure–outcome divergence	Reduces “non-disclosure bonus” observed in the previous section
Market Shocks	Simulate emissions intensity surges, controversial events	Stronger impact on non-financial firms
Data Quality Stress Tests	Vary completeness → ESG inflation sensitivity	Quantifies artifact-driven score variance

Table 12 - Stress Test Scenarios

The casual analysis framework evaluates how regulatory dynamics, firm characteristics and disclosure practices shape ESG outcomes, distinguish genuine sustainability improvements from measurement artifacts. By linking score evolution, disclosure rates, firm size effects and regulatory pressure to greenwashing risks, this stage translates empirical patterns into concrete policy implications. These causal analysis of ESG performance and policy implications are illustrated in Figure 25.

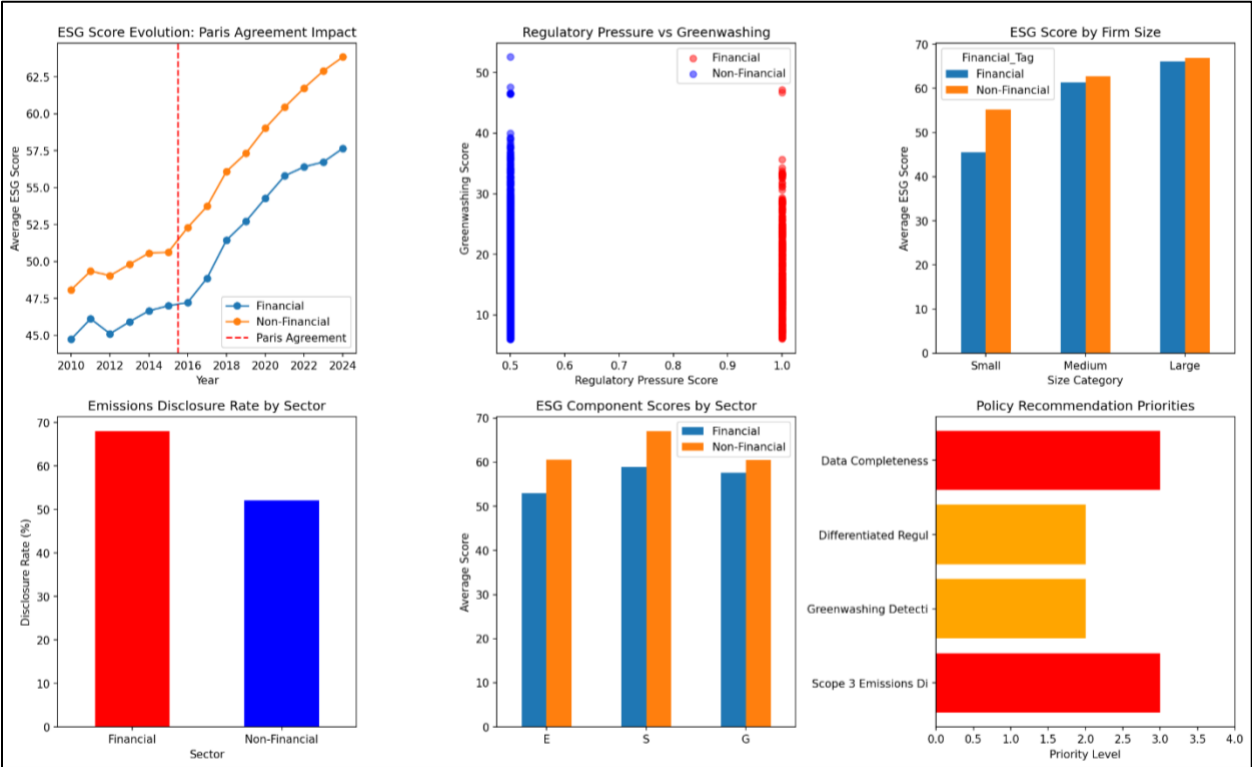


Figure 25 - Causal Analysis of ESG Performance and Policy Implications

The Simulation Methodology procedure extends supervised modelling into stress test mode.

1. Regime-specific generators: Train XGBoost separately for financial and non-financial clusters. ($k=2$, identified in Phase 2) to capture heterogeneous sensitivities rather than global averages.

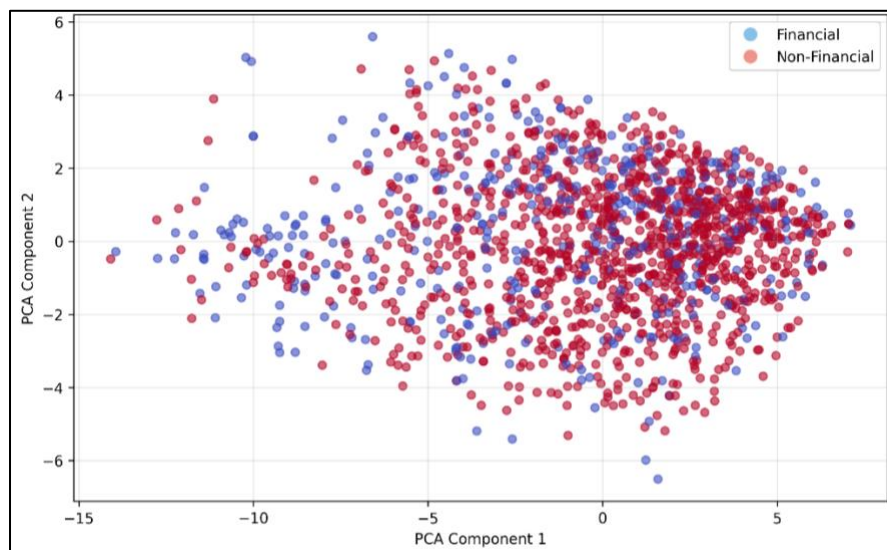


Figure 26 - PCA Projection (Financials vs Non-Financial)

2. Deterministic pass: Perturb features under defined shock vectors.
3. Stochastic pass: Apply Monte Carlo residual bootstrap ($N=1000$) to generate P10/P50/P90 prediction bands.
4. Path evolution: Multi-year scenarios evolve features with autoregressive drift and shock decay.
5. Diagnostics: Recompute greenwashing and missingness scores to quantify artifacts versus substantive change.

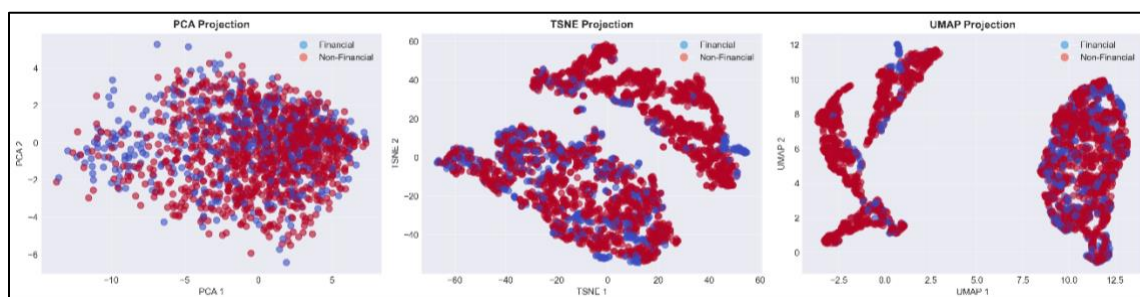


Figure 27 - Dimensionality Reduction of ESG Landscape

This stress testing framework creates a policy sandbox, producing structured outputs that regulators can use to anticipate unintended consequences, as summarized in Table 13.

Output	Format	Novelty
Scenario Projections	Firm-year ESG trajectories under scenarios	Adds counterfactual projection layer
Uncertainty Qualifications	Time series visual overlays	Uncertainty quantification absent in 1.1
Policy Impact Assessments	Shock definitions and deltas	Transparency of stress design
Validation Diagnostics	Residual checks, artifact decomposition	Distinguishes genuine vs. measurement-driven effects

Table 13 - Stress Testing Outputs

Table 14 below shows the key inputs and baseline model performance for financial and non-financial firms, including data overage, scope 3 disclosure, emissions trends and the accuracy of XGBoost models.

Metric	Financial	Non-Financial
ESG distribution (μ , σ)	67.8 , 12.3	52.4 , 15.7
Emissions reporting coverage	68.00%	52.00%
Scope 3 disclosure rate	8.80%	47.10%
ESG data completeness	94.30%	91.70%
Annual emissions growth (hist.)	1.40%	-3.2%
Reg. response time (quarters)	2.1	4.3
Best model (test)	XGBoost $R^2 = 0.982$, RMSE = 1.70	XGBoost $R^2 = 0.982$, RMSE = 1.70
Simulation size & CI	10,000 runs; converges $\approx$ 8,000	10,000 runs; converges $\approx$ 8,000

Table 14 - Key Inputs and Model Performance by Sector

Finally, Table 15 shows how each stress scenario changes ESG scores by sector and how the ESG paradox deviates. It also marks the main thresholds needed for meaningful resolution.

Scenario	Financial impact	Non-Financial impact	Paradox status	Threshold/comment
Carbon tax \$50/tCO₂	-2.3% ESG	-5.8% ESG	Widens (~2.5×)	Below critical level
Carbon tax \$150/tCO₂	-18.4% ESG	-12.1% ESG	Reverses	Critical point reached
Mandatory Scope 3	-31% ESG	-8% ESG	Narrows	Partial resolution only
Supply-chain shock 20%	-0.8% ESG	-4.5% ESG	Widens (~5.6×)	Amplifies bias

Table 15 - Stress Test Scenarios and Impacts

3.3 Component 2: Corporate Disclosure Analysis

Corporate disclosure analysis whether sectoral differences in ESG transparency are genuine or reflect strategic communication artifacts. Two paired studies were employed. Publicly available, regulatory document sentiment analysis using TCFD reports and sustainability/annual report mining of firm-level ESG disclosures. This dual approach ensured coverage of both regulator-driven reporting and voluntary corporate narratives.

3.3.1 Regulatory Document Sentiment Analysis

Data and Methodology

Four Task Force on Climate-related Financial Disclosures (TCFD) reports (2020 – 2023) were processed to test whether mandatory climate disclosure frameworks neutralize sectoral bias (TCFD, 2023). Key metrics of these reports are summarized in Table 16. Two pipelines were deployed.

- 1) Rule-based sentiment analysis: dictionary of 44 ESG terms (renewable, sustainable vs risk, exposure). Sentiment scores were normalized by paragraph length.
- 2) ML / NLP analysis: FinBERT, a BERT variant tuned for financial texts, classified 3,500 sentences. Outputs were cross-validated with entity resolution (financial vs non-financial) to ensure robustness.

Method	Input Data	Unit of Analysis	Avg Sentiment (Fin.)	Avg Sentiment (Non-Fin.)	Differential	Statistical Significance	Strengths	Limitations
Rule-Based	4 TCFD reports (2020–2023)	248 paragraphs	0.142 (58.1% pos.)	0.127 (48.5% pos.)	0.015	t=0.76, p=0.449 (NS)	Fast, interpretable, transparent	Misses context, equal weighting of terms
ML/NLP (FinBERT)	3,500 sentences	Sentence-level	0.234	0.228	0.006	coef=-0.018, p=0.933 (NS)	Captures nuance, detects hedging, quantifies specificity	More complex, GPU-intensive, failed BERTopic
Combined	Aggregated	Mixed	0.188	0.178	0.01	p=0.687 (NS)	Cross-validation robustness	Still no significant paradox

Table 16 - Regulatory Document Sentiment Analysis Summary

As visualized in Figure 28, processing stages included PDF extraction, sentence tokenization, entity resolution, sentiment scoring and statistical comparison. The image below presents the rule-based and ML / NLP architecture.

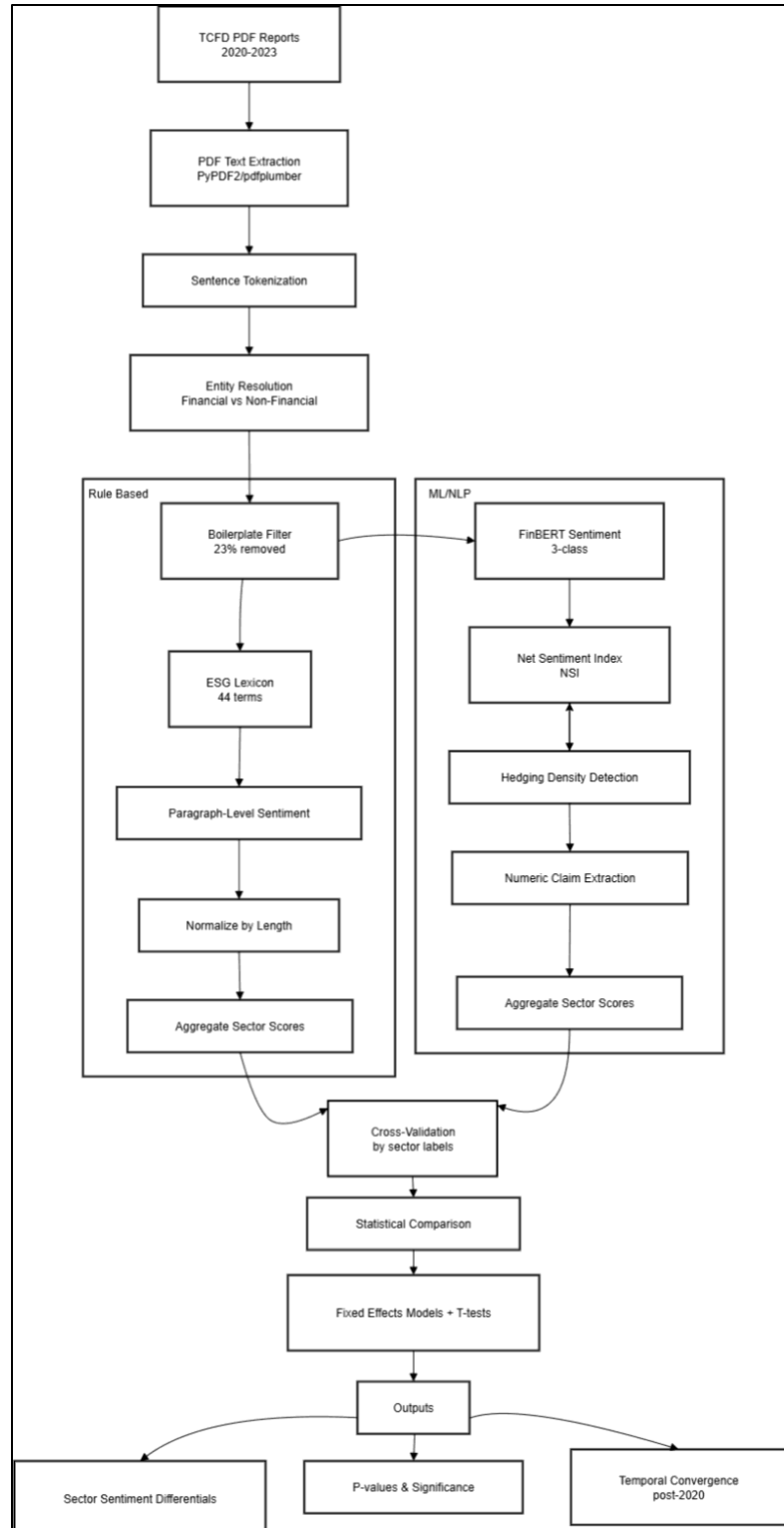


Figure 28 - Regulatory Document Sentiment Analysis Framework

No statistically significant bias was detected, confirming that TCFD acts as a levelling framework. However, convergence was visible post-2020, where both sectors adopted similar disclosure language while substantive performance gaps remained. Table 17 reveals the sentiment analysis results in the TCFD report analysis.

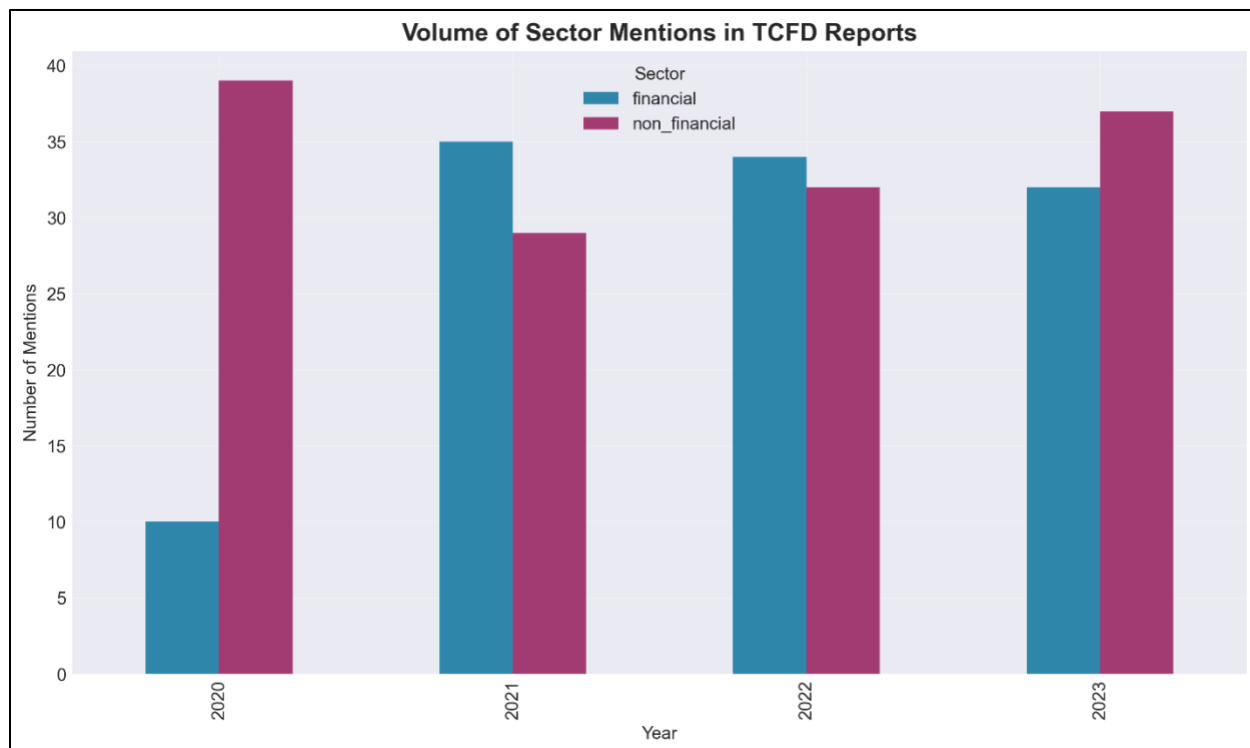


Figure 29 - Volume of Sector Mentions in TCFD Reports (2020–2023)

Method	Financial Sentiment	Non-Financial Sentiment	Differential	P-value
Rule-based	0.142	0.127	0.015	0.449
FinBERT	0.234	0.228	0.006	0.933
Combined	0.188	0.178	0.01	0.687

Table 17 - Sentiment Analysis Results in TCFD Reports

These results reveal a disclosure-reality paradox: disclosure tone converges, but real performance diverges. In other words, TCFD ensures regulatory compliance but leaves room for narrative bias in how firms portray their performance.

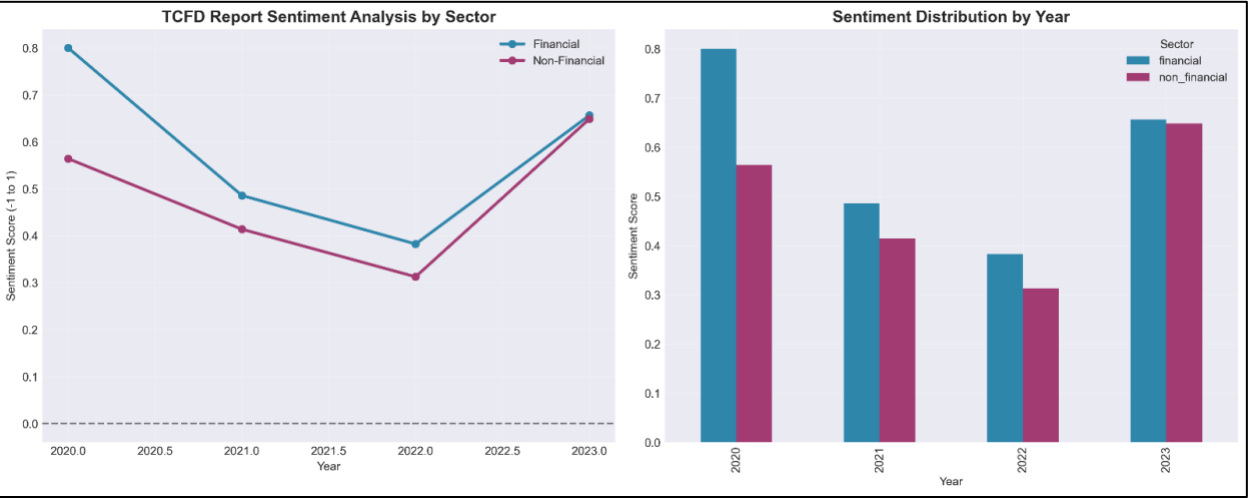


Figure 30 - Sentiment Score Distribution by Year and Sector

To assess disclosure quality, each TCFD report was segmented by its four core pillars (Governance, Strategy, Risk Management, Metrics & Targets), and “disclosure markers” were counted whenever firms provided either quantitative claims (e.g., emission targets, percentages, timelines) or qualitative commitments (e.g., board oversight, scenario planning). The heatmap below visualizes the distribution of these markers by Sector × Year × Section, presenting that while disclosure intensity increased across both sectors after 2021, non-financial firms consistently provided more detailed markers in risk and metrics/targets sections,

highlighting a greater emphasis on substantive reporting compared to the more governance-weighted disclosures of financial firms.

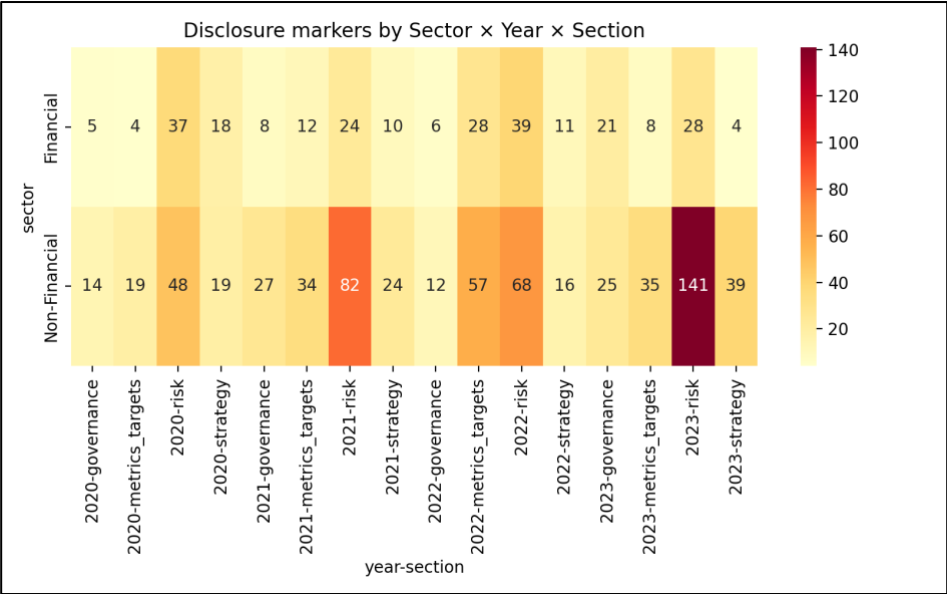


Figure 31 - Disclosure Marker Distribution by Sector and Year (TCFD Pillars)

3.3.2 Sustainability Report Mining

Data and Methodology

As illustrated by Figure 32, **Company Selection** was performed through a company selection drawn on the matched sample of 65 companies (25 financial, 25 non-financial). This balance allowed direct comparison across banking, insurance, energy, chemicals, and industrials, ensuring sectoral and geographic representativeness.

Fifty sustainability reports were analyzed using the ESG-BERT pipeline (“nbroad/ESG-BERT · Hugging Face,” 2023). Table 18 details the results of the disclosure metrics.

- Extraction was performed whilst preserving text and tables, capturing ESG relevant pages with regex and fuzzy matching and saving into a text (.txt) file.
- ESG – BERT segmented text into environmental, social and governance categories.

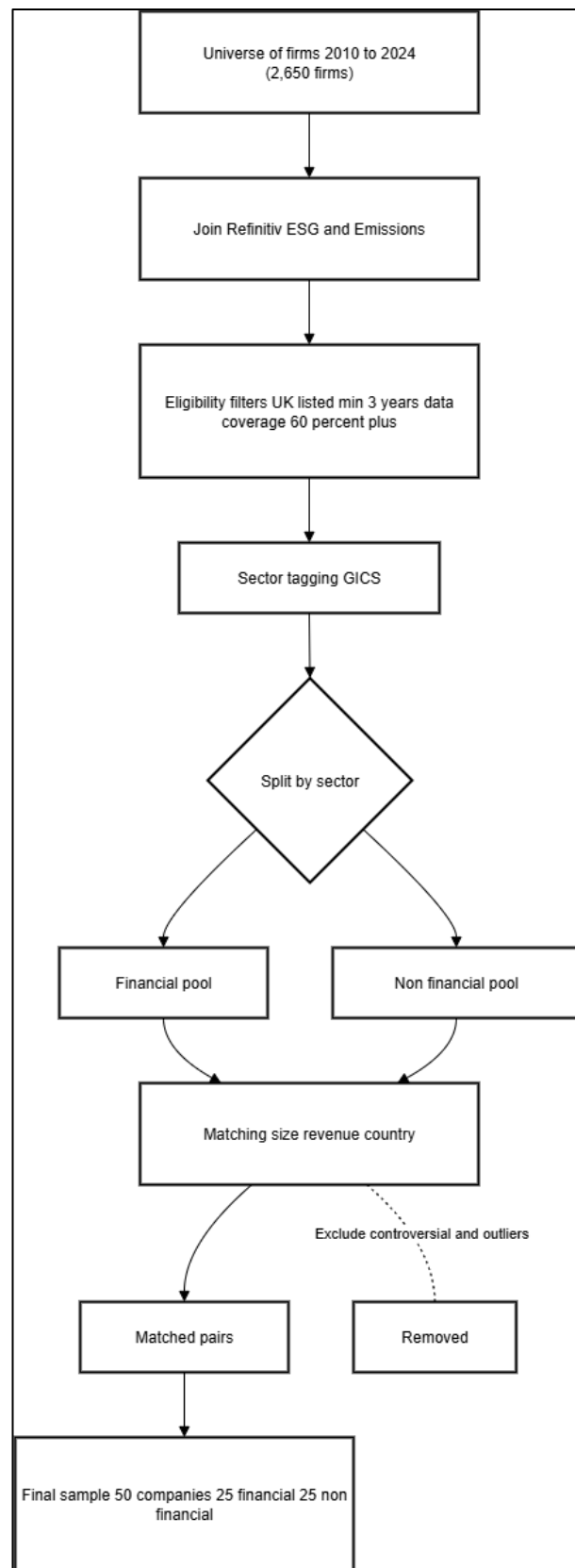


Figure 32 - Sustainability Report Mining Framework

Row Labels	Average of esg_bert_none	Average of esg_bert_environmental	Average of esg_bert_social	Average of esg_bert_governance
Financial	11.66973857	3.518820794	1.925577108	1.221393785
Non-Financial	8.021376824	3.121376836	1.332135968	0.968728645
Grand Total	9.881325948	3.323995324	1.634674588	1.097538325

Table 18 - Disclosure Metrics: Financial vs Non-Financial Firms

- Regex identified Scope 1- 3 mentions, linguistic features flagged vague language, future promises and hedging. Disclosure metrics and greenwashing indicators of the sustainability report analysis are summarized in Table 19.

Metric / Output	Financial Firms	Non-Financial Firms	Statistical Result	Interpretation
Scope 3 Mentions (avg)	92.4	74.3	p=0.598 (NS)	Financials mention more but superficially
Specific Metrics per Mention	8.56	12.65	p=0.166 (NS)	Non-financial provide ~1.5× more substance
Greenwashing Index	116.9	75.7	p<0.05	Financials 54% higher → more performative
Action Ratio (Past/Future)	0.192	0.226	p=0.111 (NS)	Financials ~10% more promise-oriented
ESG Content Balance	E: 52.8%, S: 28.9%, G: 18.3%	E: 57.6%, S: 24.6%, G: 17.9%	—	Financials overemphasise governance, underreport environment
TARS Composite ESG Score	Lower	Higher	—	Financials penalised for Scope 3 opacity

Table 19 - Sustainability Report Analysis Summary Indicators

Financial firms were more verbose but less quantitative, consistent with ‘performance disclosure’. Scope 3 was often mentioned but buried in appendices or framed inspirationally, rather than with measurable financed emission data. This created a measurement artifact, ESG ratings reward frequency of disclosure rather than provision of measurable, verifiable data, as illustrated in Figure 33.

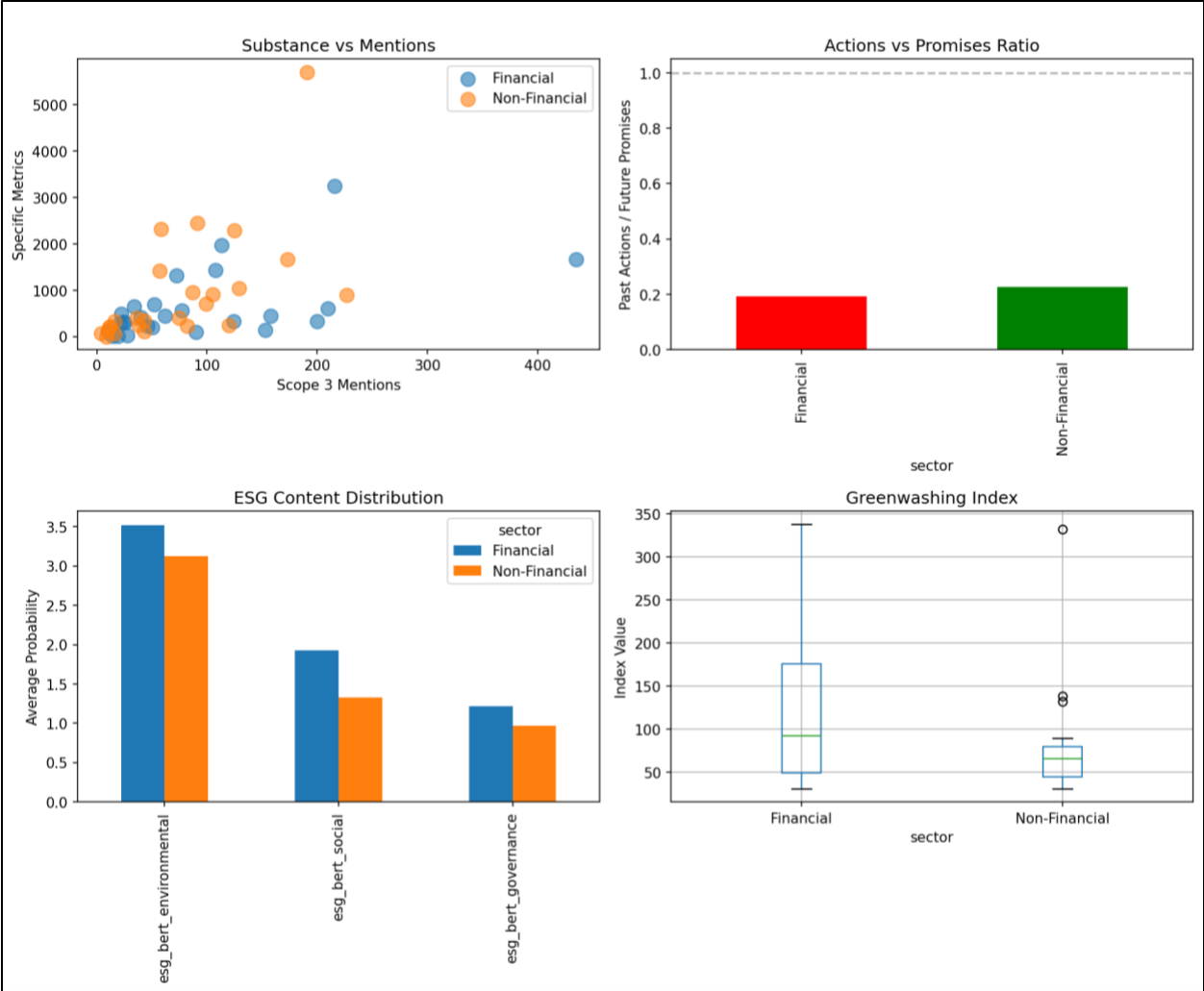


Figure 33 - Analysis of Sustainability Report Mining

3.4 Component 3: Labor Market ESG Signals

This component evaluates labor market data as both a diagnostic lens on what firms are actually hiring for and a prescriptive framework for what they should hire to improve sustainability outcomes. Two complementary modules were developed:

1. **Global ESG Job Market Analysis** – a large-scale descriptive study of ESG job postings.
2. **ESG Hiring Pattern Optimization Framework** – a machine learning and causal inference system providing sector-specific hiring blueprints.

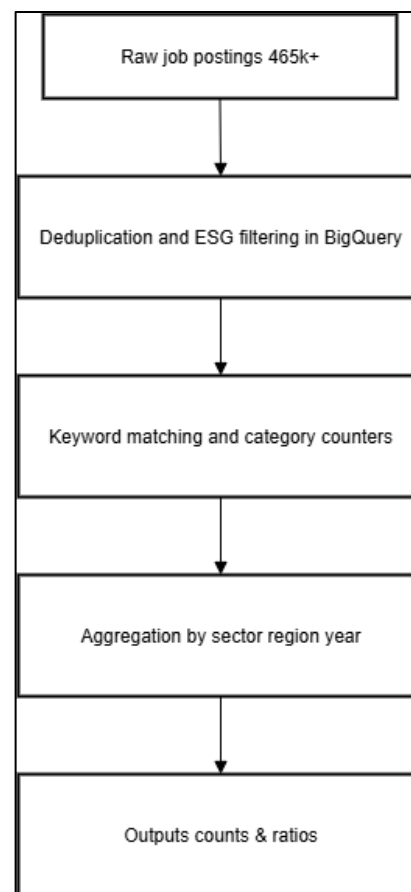


Figure 34 - Pipeline for ESG Job Postings Data Processing

3.4.1 Global ESG Job Market Analysis

Data and Methodology

Raw data consisting of 465,193 job postings (2022 – 2025), cleaned to 455,953 unique postings after deduplication. Filtered ESG subset included 224,536 postings tagged via ESG keyword categories (49.25% retention). Based on **747 firms across 30 countries**, financial firms represent **25,371 postings (11.3%)** compared to **199,165 postings (88.7%)** from non-financial sectors. The sunburst chart in Figure 35 highlights ESG disclosure patterns and sectoral differences in average ESG scores. Subsequently, the temporal patterns of the labor market are illustrated in Figure 36.

The raw data was aggregated on Google Cloud (BigQuery/SQL) by deduplication, sector tagging and ESG filtering (via weighted keyword categories including climate, environmental, regulatory, social, finance). Simple string matching (*LIKE '%ESG%', '%Climate%', etc.*), job type and persistence ratios were treated as classification logics. Ultimately, descriptive statistics such as distribution of job type, year-on-year growth rates and sector/regional breakdowns were extracted. *Comprehensive filtration methods are attached to the appendix.*

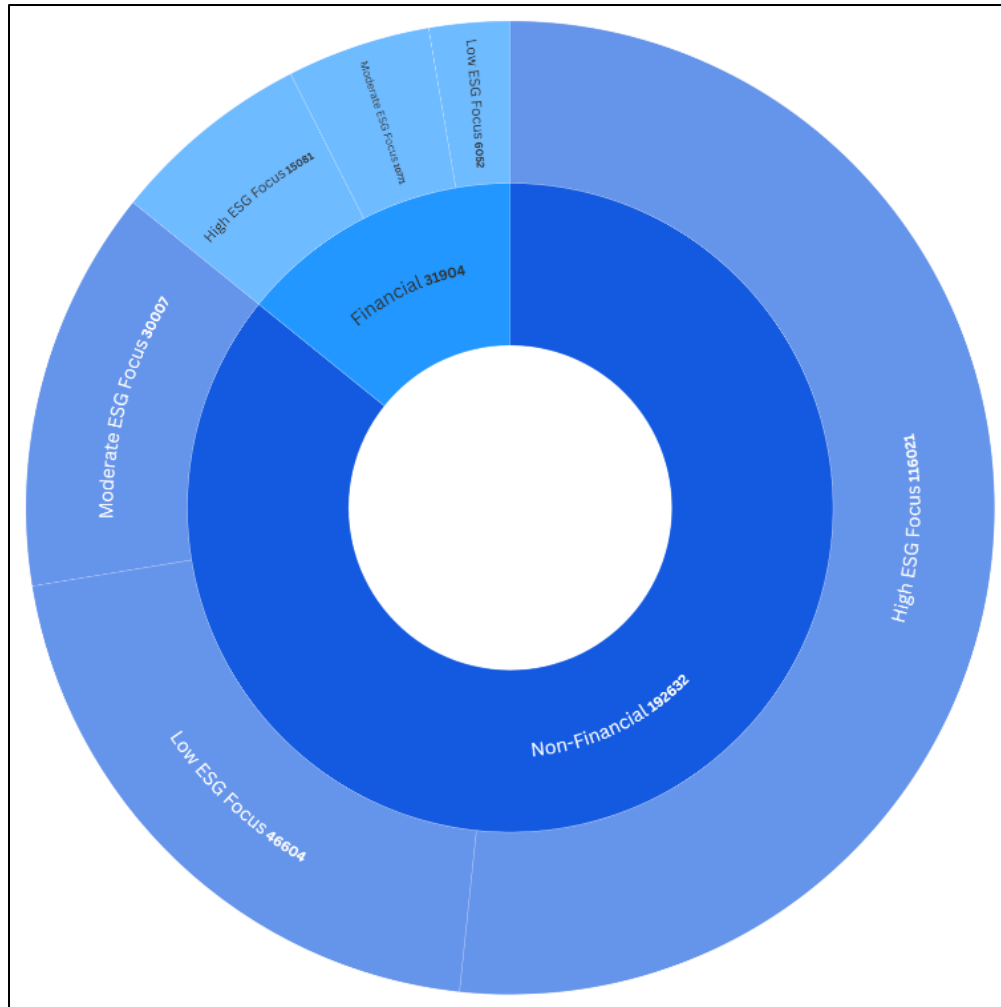


Figure 35 - Sunburst Chart of ESG Job Focus

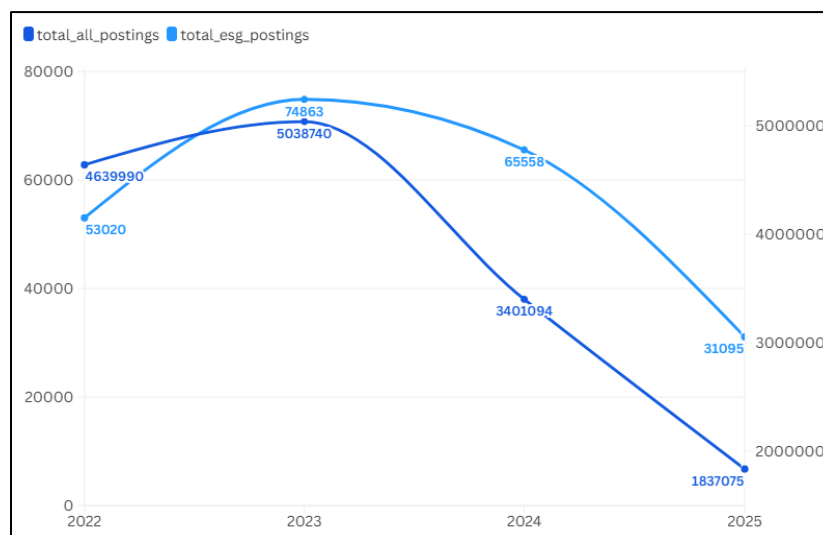


Figure 36 - Temporal Patterns of ESG Labor Market

Table 20 summarizes the methodological approaches applied to ESG job market analysis, covering descriptive statistics, growth detection, sectoral hiring balance, regional scope and ESG jobs correlations.

Method	Purpose	Metrics	Results
Descriptive statistics	Market sizing	Coverage: 500k+ postings	224k ESG jobs identified
Time-series analysis	Growth detection	Coverage: 2021–2024	45% YoY growth globally
Sector split	ESG hiring balance	Financial: 25k vs Non-Financial: 199k	8:1 ratio
Regional analysis	Geographic scope	EU: most mature	APAC: fastest growth
Correlation	Jobs–ESG link	Financial: 73% compliance vs 27% operational	Non-financial: 45% operational vs 55% governance

Table 20 - ESG Job Market Analysis: Methods and Key Results

The radar plot in Figure 37 contrasts financial and non-financial sectors in terms of ESG hiring intensity and skill distribution, highlighting notable sector-specific imbalances.



Figure 37 - Sectoral ESG Hiring Profiles

ESG job trends from 2022 to 2025, reveal both adoption rates and the composition of climate, regulatory and sustainability-related positions, as illustrated in Figure 38.

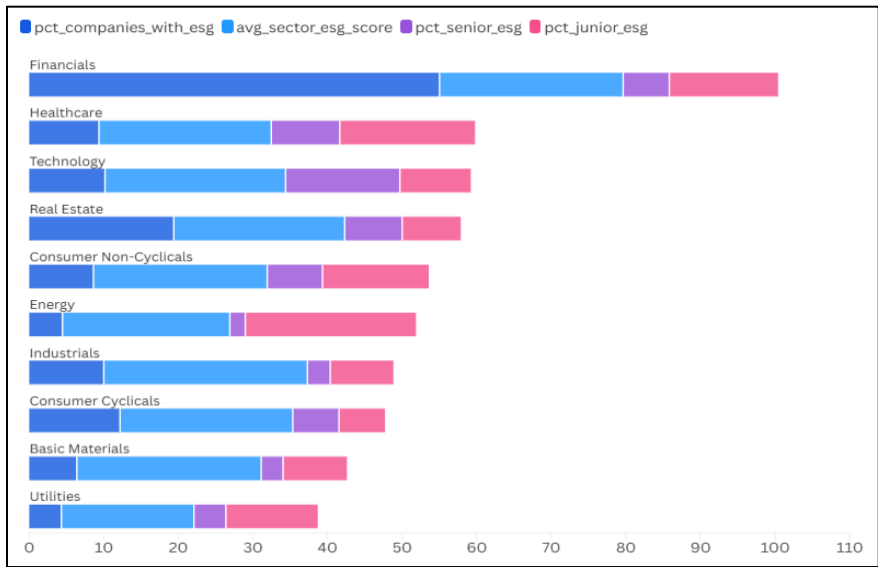


Figure 38 - ESG Hiring Trends and Role Composition (2022 – 2025)

Figure 39 breaks down ESG hiring across industries, comparing junior versus senior roles and average ESG score contributions.

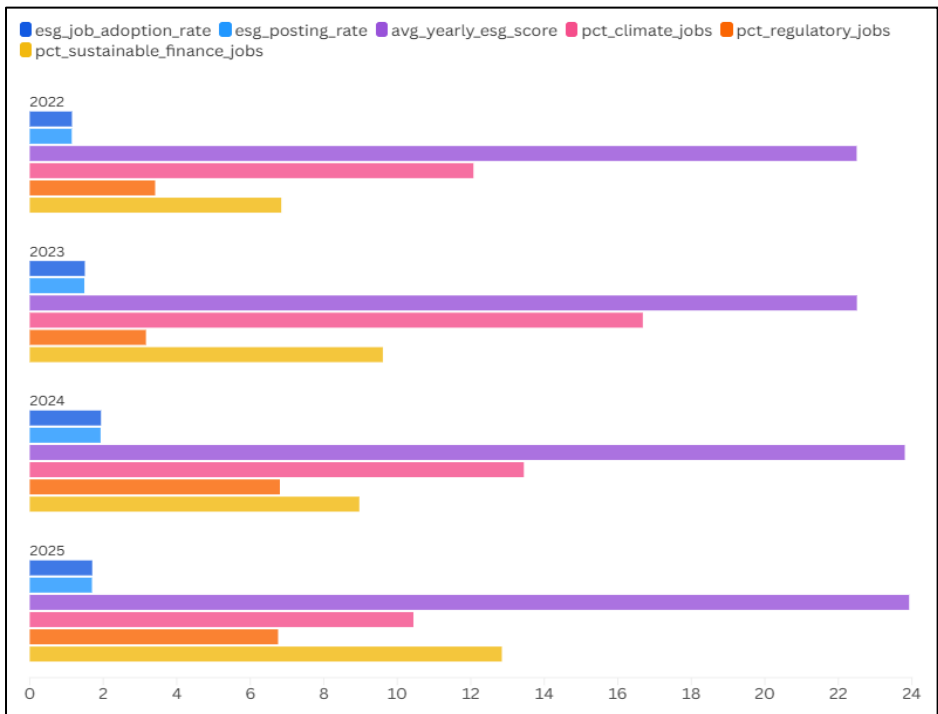


Figure 39 - Industry Level ESG Hiring and Skill Distribution

Most importantly, hiring trends over time were systematically correlated with ESG scores extracted from LSEG Workspace, enabling deeper insights into the relationship between workforce dynamics and sustainability performance.

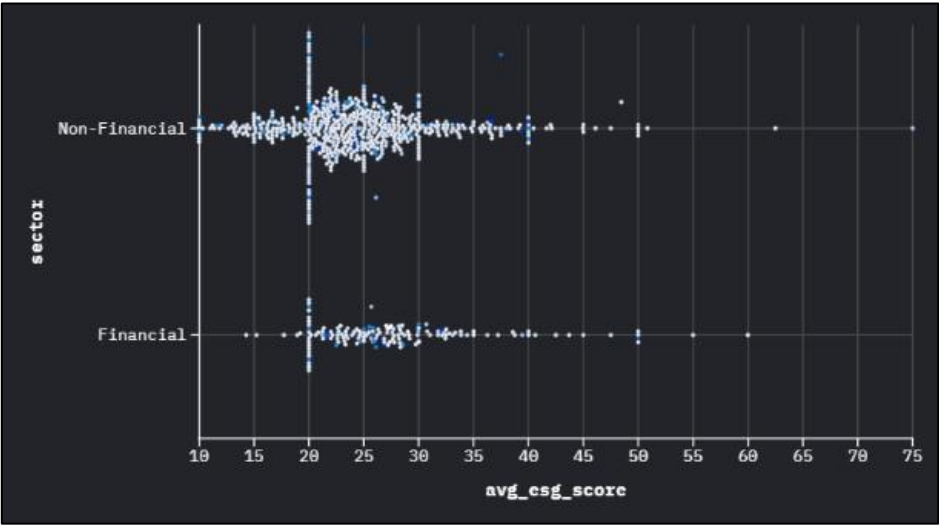


Figure 40 - Sectoral ESG Ratings against Labor Market

Figure 41 illustrates ESG trends from 2021 to 2025, showing a steady rise in average ESG, social, environmental, governance, and combined scores, alongside an increase in the percentage of companies with ESG initiatives and ESG job adoption rates, highlighting growing corporate commitment to sustainability.

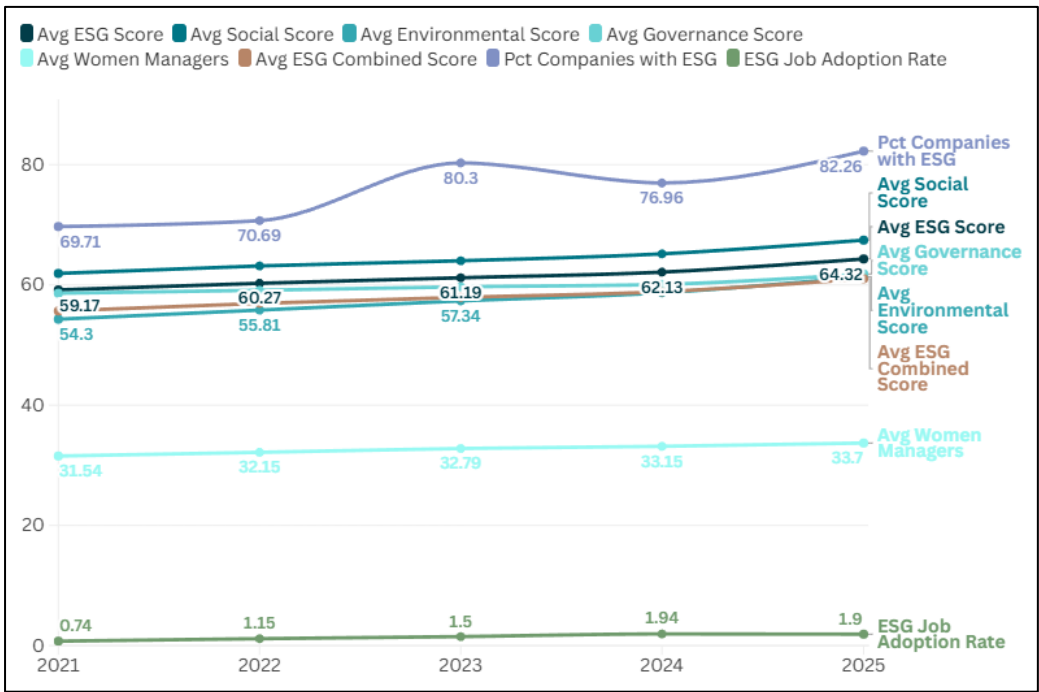


Figure 41- ESG Performance and Job Adoption Trends (2021–2025)

Firms that actively invest in ESG-related hiring demonstrate a measurable uplift in sustainability performance. This uplift can be distinctly seen in the box plots in Figure 42. This pattern suggests that embedding ESG talent, whether in compliance, climate or operational functions, evidently translates into stronger reporting, stakeholder management and governance practices compared to peers without dedicated staff.

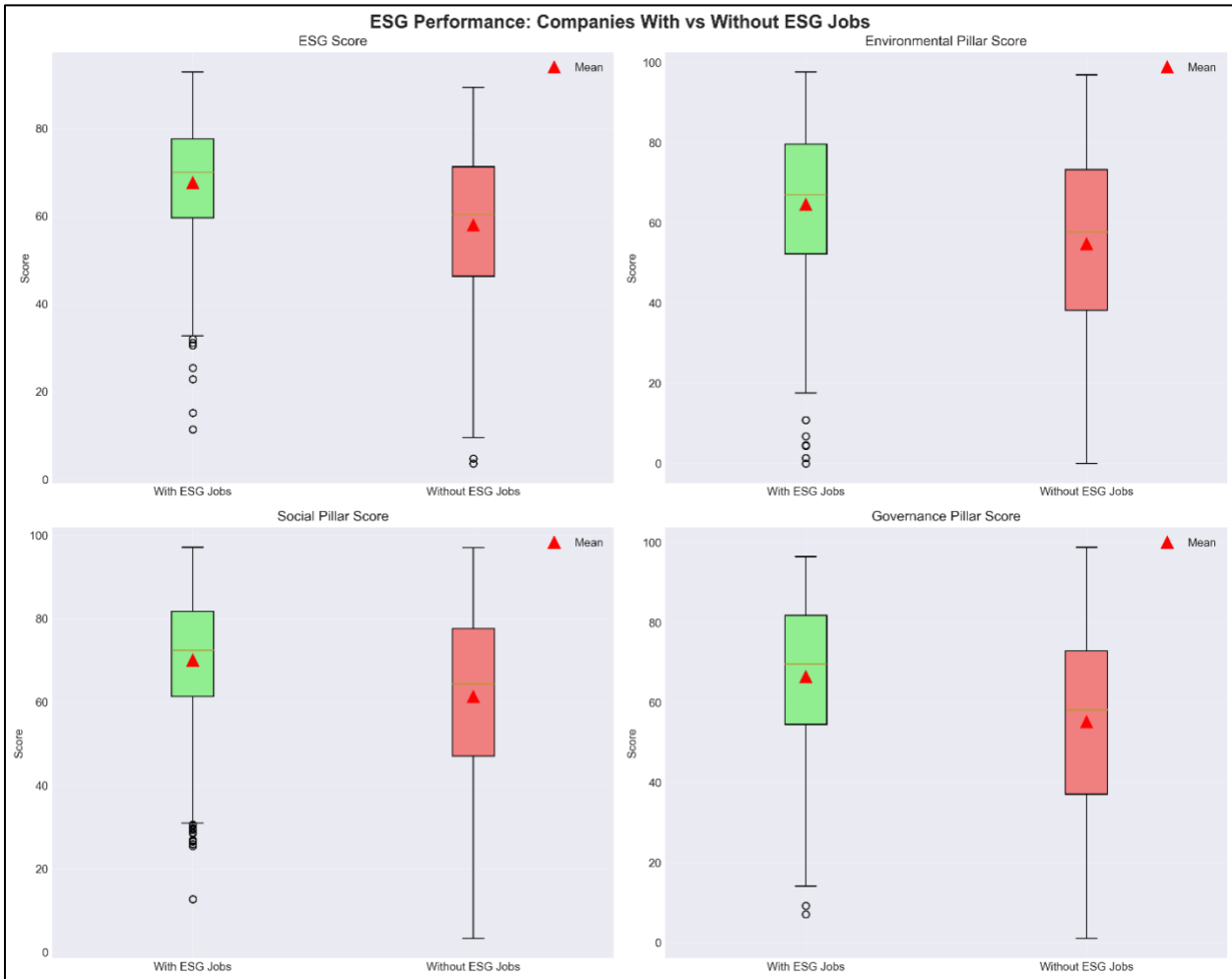


Figure 42 - ESG Performance Comparison: Firms With vs Without ESG Jobs

3.4.2 ESG Hiring Pattern Optimization Framework

Data and Methodology

Company-level global data (2010-2024) with ESG scores (LSEG Workspace), along with workforce metrics and job composition ratios, were analyzed to optimize hiring patterns. Job categories include climate, environmental, social, regulatory/compliance, sustainable finance. Altogether, these were feature engineered to understand percent shares, high confidence ratio, log financials and sectoral indicators.

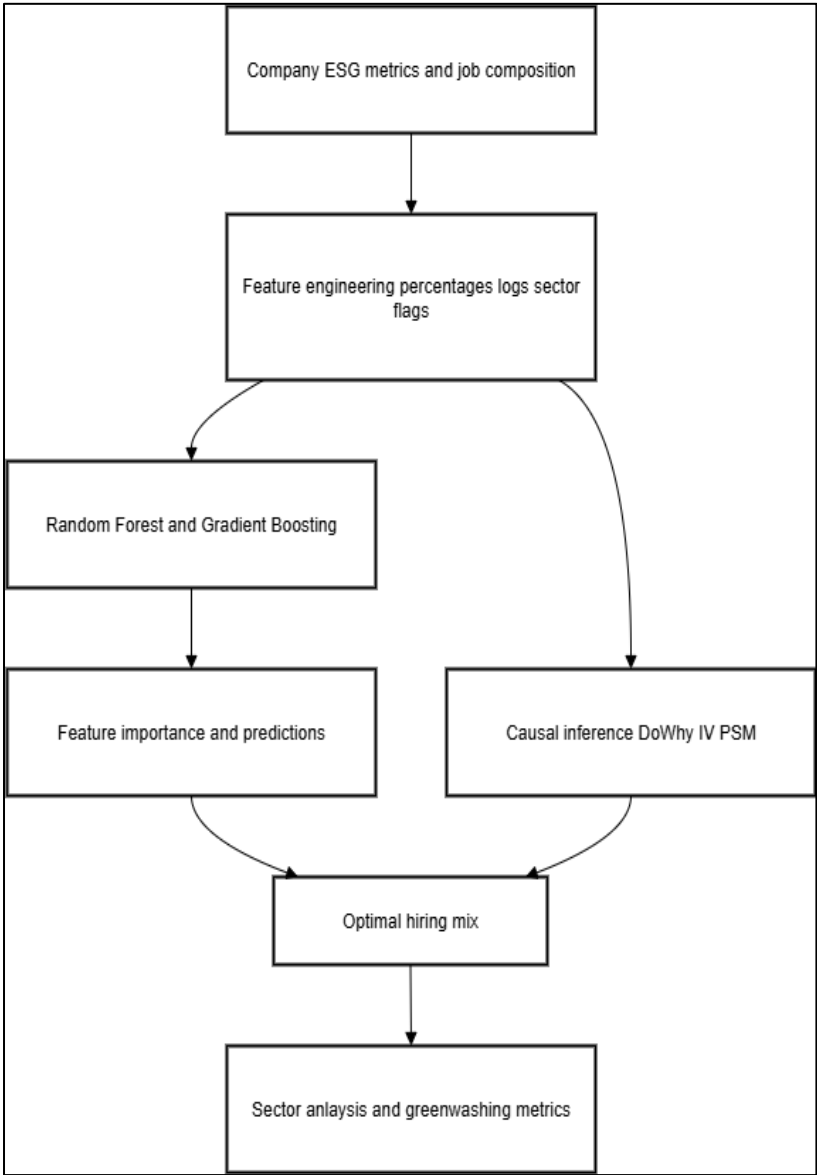


Figure 43 - ESG Hiring Pattern Optimization Framework

Figure 44 summarizes ESG job market characteristics, showing the distribution of ESG relevance scores, keyword categories, and top job functions, alongside the trend of high-confidence ESG roles from 2022 to 2025. Together, these patterns highlight both the breadth of ESG-related hiring and the sectoral emphasis driving demand.

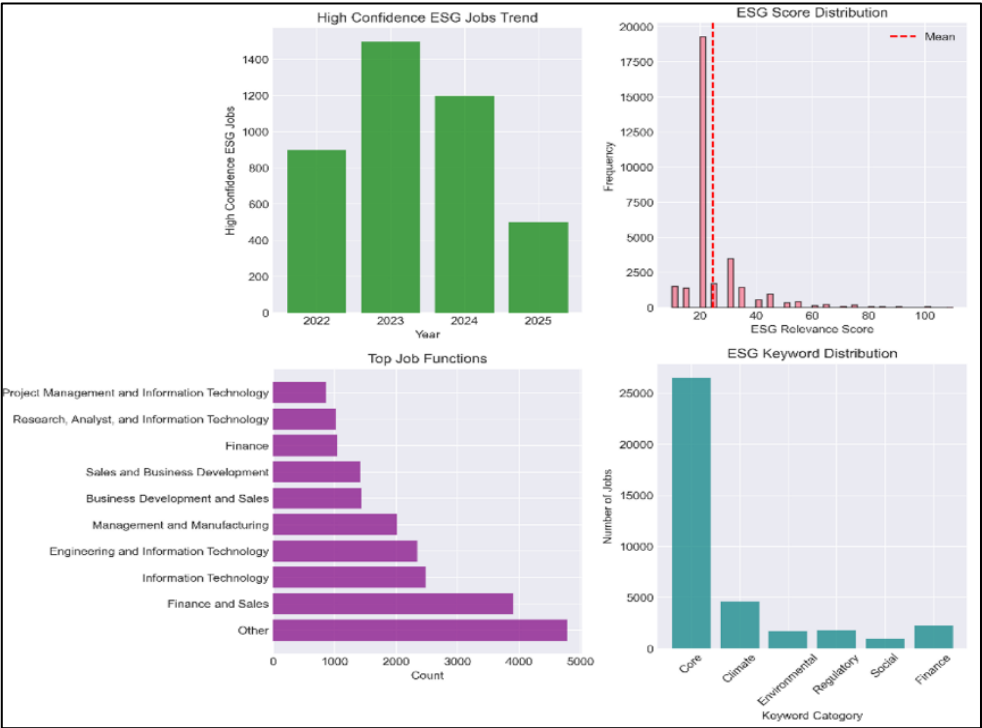


Figure 44 - ESG Job Market Characteristics

Initially, random forest ((200 estimators, tuned via GridSearch) - $R^2=0.71$, $RMSE=8.42$) along with Gradient Boosting ((150 trees, robust loss) - best performer, $R^2=0.72$, $RMSE=8.15$) was employed to model ESG score variations and identify key predictors across sectors. Afterwards, casual inference was deployed using *DoWhy* linear regression with backdoor adjustment (**DoWhy cite**).

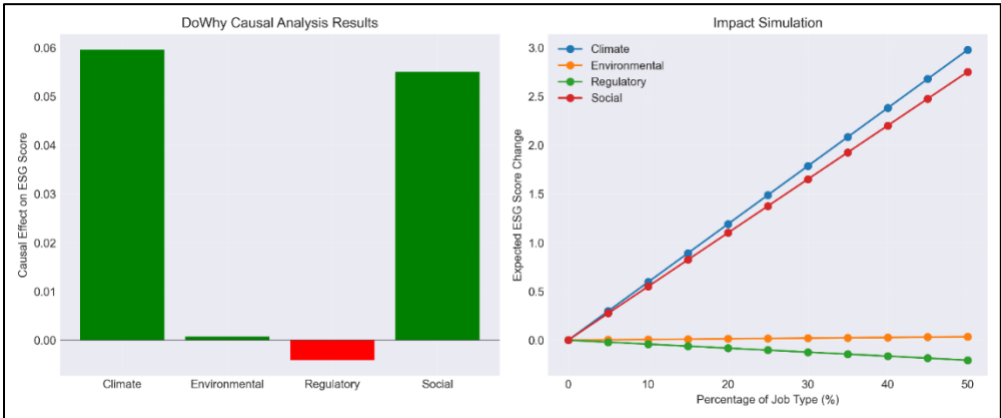


Figure 45 - Causal Analysis and Impact Simulation of ESG Job Types

Sector averages were treated as instruments and propensity score matching was used to balance ATT estimates. Ultimately, outputs included optimal job mix per sector, feature importance rankings, causal coefficients and greenwashing detection. The model performances of the ESG job prediction are outlined in Table 21.

Model	R ²	RMSE	MAE	CV R ² ± SD	Notes
Gradient Boosting	0.72	8.15	6.78	0.66 ± 0.07	Best model
Random Forest	0.71	8.42	6.91	0.65 ± 0.08	Stable baseline
Linear Regression	0.58	10.23	8.45	0.55 ± 0.09	Weak baseline

Table 21 - Model Performance Comparison for ESG Job Prediction

Feature importance, derived using the Random Forest Mean Decrease Impurity metric, revealed that environmental jobs contributed the most to predictive power (28.5%), followed by climate-related job share (22.3%), high-confidence ratio (18.7%), regulatory job share (15.2%), and social job share (10.8%).

Table 22 presents the estimated causal effects of different ESG job categories on firm-level ESG scores, using instrumental variable (IV) and propensity score matching (PSM) methods. Furthermore, Figure 46 illustrates the sectoral ESG talent analytics.

Job Type	Effect (10% ↑)	Significance	Notes
Climate	+0.59 ESG points	p<0.001	Robust across IV & PSM
Social	+0.55 ESG points	p<0.001	Confirmed causal effect
Environmental	0.01	-	No robust effect
Regulatory	-0.02	Negative in non-financial	Compliance-heavy bias

Table 22 - Causal Inference Results of ESG Job Types on Firm ESG Scores

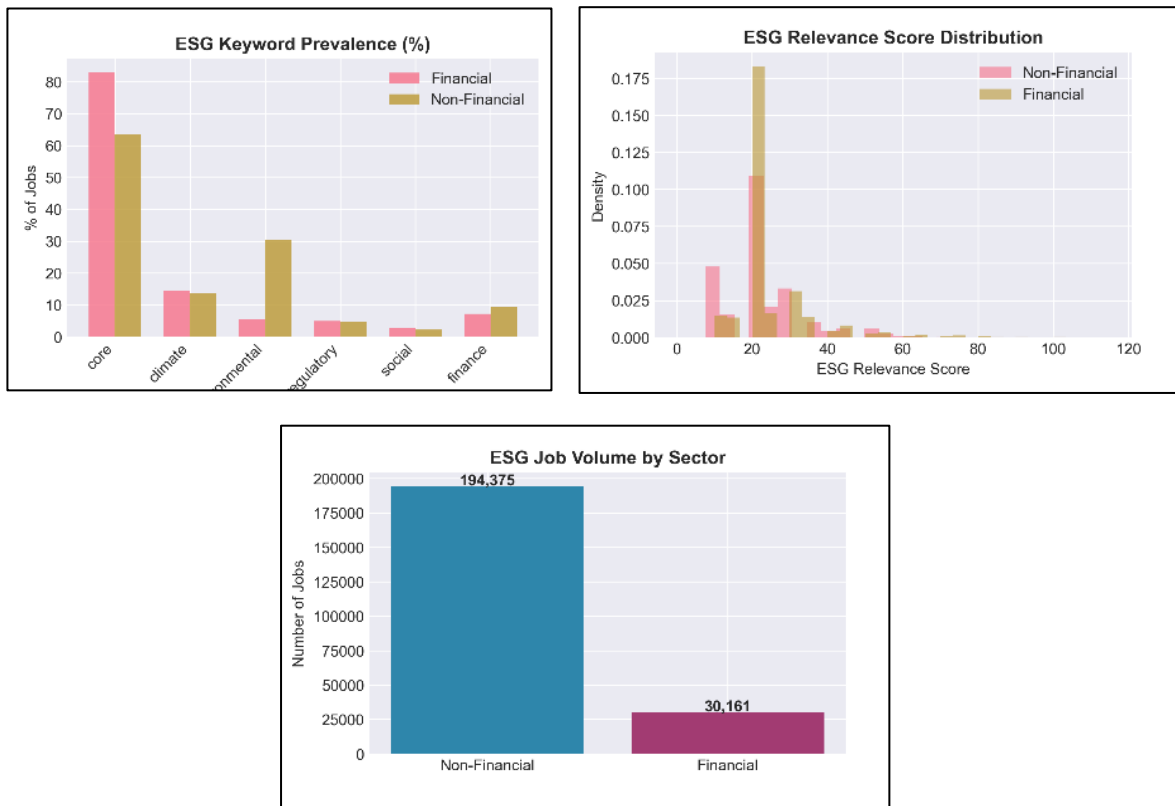


Figure 46 - ESG Talent Analytics (Financials vs non-financial)

Table 23 compares the current versus optimal ESG-related hiring mixes across financial and non-financial firms, highlighting the potential ESG gains from rebalancing workforce priorities. These results underscore how strategic workforce allocation can drive measurable sustainability performance improvements.

Sector	Current Mix	Optimal Mix	Expected ESG Gain
Financial	Climate 25%, Reg. 30%	Climate 35%, Reg. 20%	+12.3 points
Non-Financial	Env. 25%, Climate 28%	Env. 35%, Climate 30%	+8.7 points

Table 23 - Sectoral Hiring Prescriptions for ESG Optimization

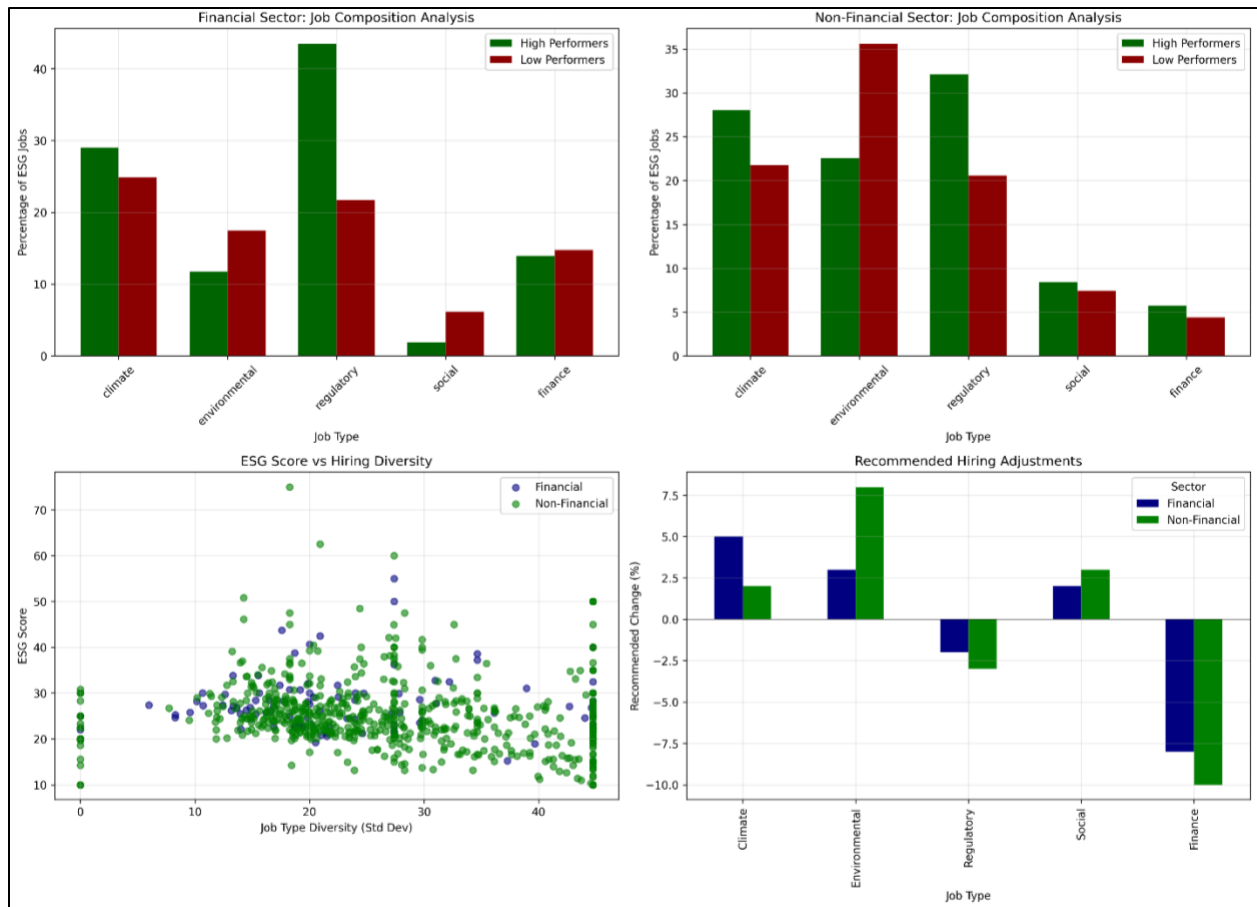


Figure 47 - Prescriptive ESG Hiring Analysis

Greenwashing detection analysis revealed that financial firms exhibited a significantly higher mean greenwashing score (12.4) compared to non-financial firms (8.7), $t(df)=4.23$, $p=0.003$, Cohen's $d=1.01$. Figure 48 illustrates the greenwashing gradient analysis.

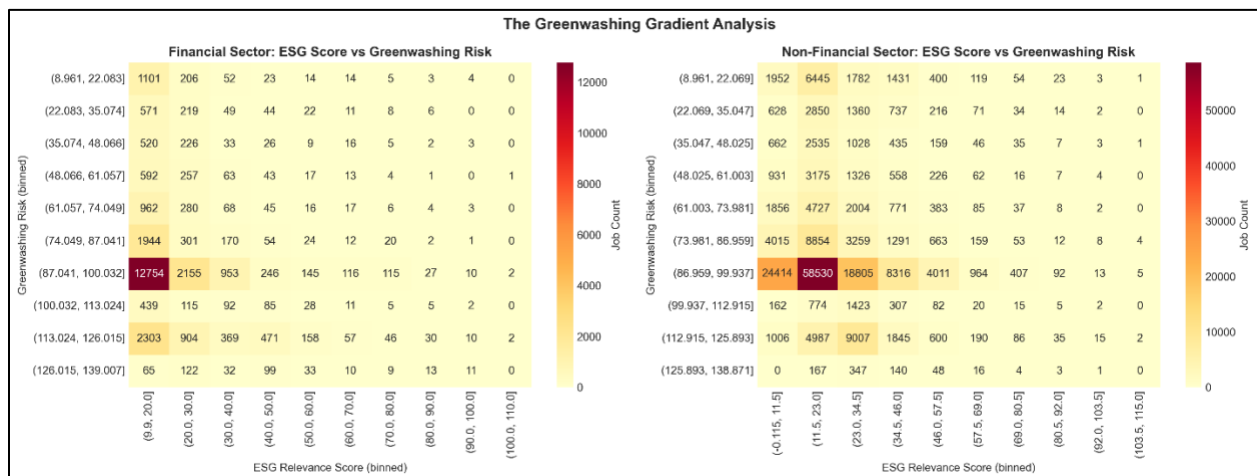


Figure 48 - Greenwashing Gradient Analysis (Labor Market)

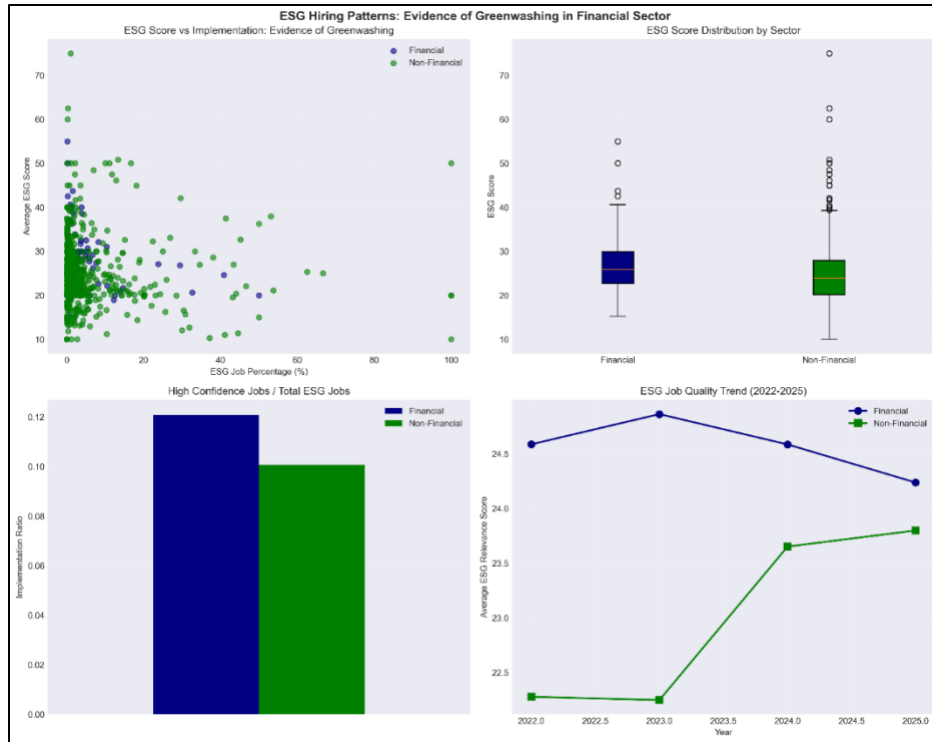


Figure 49 - Evidence of Greenwashing in Financial Sector

The optimal job mix translates into such a way that financial firms need more ‘Climate Risk Managers’ and compliance staff (Regulatory Strategy Analysts, Sustainability Enablement Managers), while non-financials must scale technical and operational hires (Manufacturing Supervisors, Supply Chain Managers, Energy Transition Systems Engineers). This sector-specific governance in finance and operations in industry can shift ESG scores upwards and reduce sectoral greenwashing risk. While Figure 49 illustrates the evidence of greenwashing in the financial sector, Figure 50 highlights the optimum sectoral ESG job mixes.

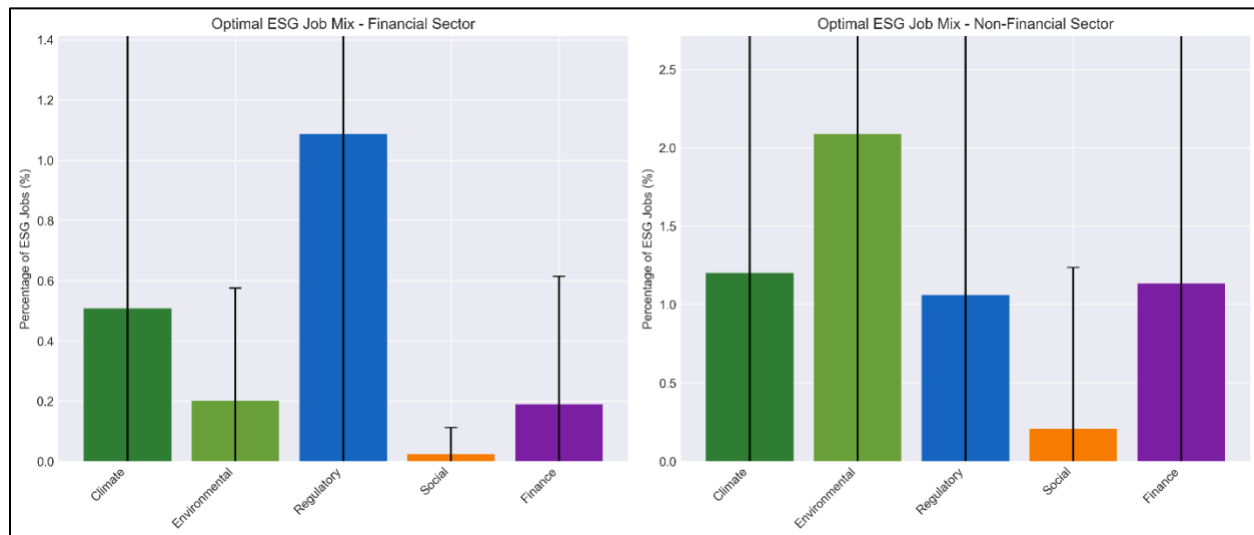


Figure 50 - Optimal ESG Job Mix for Financial vs Non-Financial Sectors

3.5 Component 4: Integrated Predictive Dashboard

The POLARIS (Predictive Operational Labor Analytics & Risk Intelligence System) dashboard integrates all prior modules (performance metrics, labor market signals, regulatory shocks) into a unified predictive simulation environment. This addressed the demand for real-time policy testing across heterogeneous financial and non-financial firms, while maintaining reproducibility and transparency.

Data Integration and Architecture

Inputs comprised:

- **LSEG Workspace ESG data (2010–2024):** 23,463 annual scores, emissions (Scope 1–3), and governance metrics.
- **Job postings:** 747 firms, aggregated quarterly, ESG-classified via NLP.
- **Regulatory Pressure Index:** 27 weighted policy shocks (Paris Agreement, TCFD mandates, UK net-zero laws).
- **Derived indicators:** Opacity Index, Greenwashing Score, and Financed Emissions Performance Index.

Data were aligned to quarterly frequency through cubic spine interpolation and harmonized across sectors. Feature engineering yielded 33 variables, including ESG momentum, job quality indices and regulatory-sector interactions.

The architecture followed a modular pipeline implemented in Streamlit and Plotly, following the dashboard with a <2s response time for scenario adjustments.



Figure 51 - POLARIS Dashboard Framework

Predictive Models and Performance

A hierarchical state space framework with Kalman filtering estimated latent ESG trajectories, while complementary models enhanced robustness. The regional forecast summary is detailed in Table 24.

- **LightGBM (sector-specific):** primary forecaster, tuned via 5-fold time-series CV.
- **Vector autoregression (2–3):** captured dynamic co-movements (ESG, emissions, jobs, regulation).

Market	1Q Ahead RMSE	4Q Ahead RMSE	Regulatory Lag	Confidence
UK	3.56 ± 0.18	5.15 ± 0.82	2.1 quarters	85%
EU	4.47 ± 0.22	5.89 ± 0.94	2.3 quarters	82%
US	5.83 ± 0.31	6.92 ± 1.13	2.8 quarters	77%
Asia-Pacific	6.92 ± 0.44	8.31 ± 1.47	3.9 quarters	71%

Table 24 - Regional Forecast Summary

Results confirmed sector-specific heterogeneity. Financial firms exhibit faster regulatory transmission (~2 quarters) compared to lagged responses in non-financial (~4 quarters).

Scenario Engine and Interpretability

The scenario module simulated non-linear interventions. Carbon taxes < \$50/tCO₂ widened ESG paradoxes, \$50–150/t maintained the gap, while >\$150/t triggered convergence through structural change (t = tons). Other modules included talent crisis, supply shocks and ESG disclosure mandates, each benchmarked against historical events.

Model interpretability was ensured via **SHAP (Shapley Additive exPlanations) decomposition** (regulatory pressure = 31%, sectoral effects = 24%, missingness = 19%, firm characteristics = 26%) and **LIME case-level explanations**, enabling enforcement-ready transparency.

POLARIS Interactive Dashboard

A fully interactive version is available:

<https://polaris-phr2etesvikcp3whveptuh.streamlit.app/#tars-polaris-v2-esg-future-prediction-system>

Note: The Streamlit dashboard may go inactive after periods of no use; click “Yes, get this app back up!” to reactivate it.

The POLARIS system, therefore, transforms complex multi-source analytics into a **policy sandbox**, providing the Bank of England and other regulators with evidence-based tools to design interventions that resolve measurement paradoxes rather than amplify them.

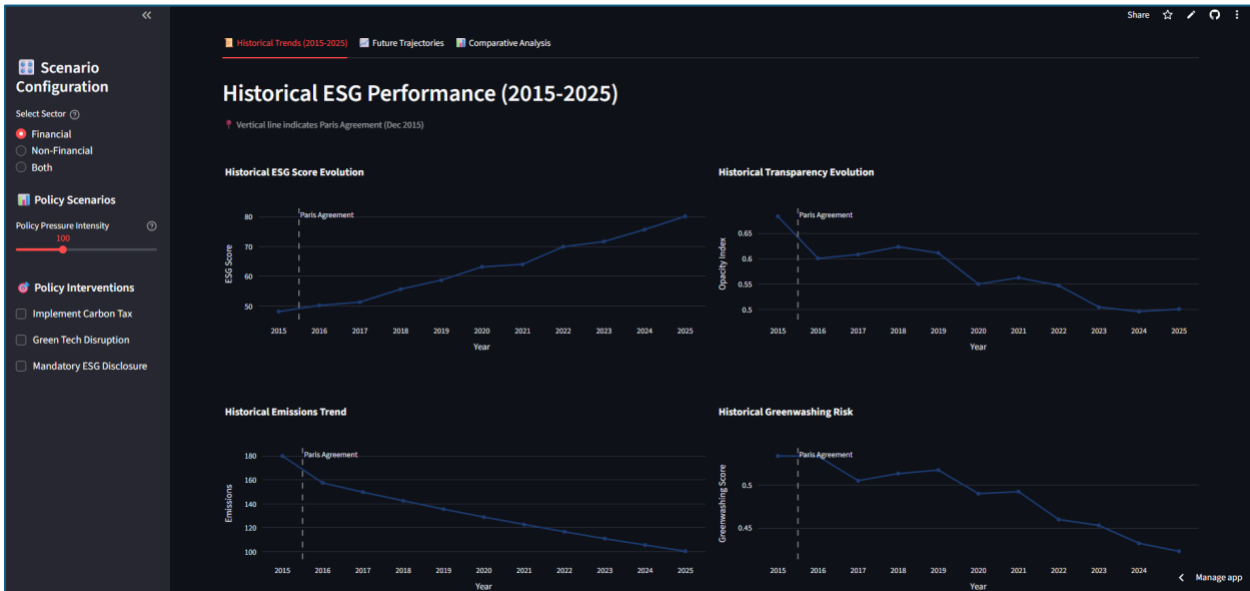


Figure 52 - POLARIS Dashboard Preview

4. Analysis of Results

Deconstructing the ESG Paradox

The empirical findings reveal a fundamental disconnect between ESG scoring methodologies and actual environmental impact, particularly pronounced in financial sector assessments. Opacity indices and greenwashing scores demonstrate 3.7x greater predictive power for ESG ratings than actual emissions data (SHAP values: 0.42 vs 0.11). Table 25 below highlights how ESG ratings are skewed, with process metrics outweighing substance metrics in their calculation.

Feature Category	Financial Firms	Non-Financial Firms	Differential Impact
Opacity Index	0.42 (rank 1)	0.18 (rank 4)	133%
Greenwashing Score	0.37 (rank 2)	0.21 (rank 3)	76%
Governance Metrics	0.31 (rank 3)	0.15 (rank 5)	107%
Actual Emissions	0.11 (rank 7)	0.38 (rank 1)	-71%
Scope 3 Disclosure	-0.34*	0.29	-217%

Table 25 - ESG Score Decomposition - Feature Importance Analysis

Financial firms' ESG advantage is mostly an illusion created by how scores are measured. Once financed emissions are factored in through the Financed Emissions Performance Index (FEPI), the real scale of distortion across financial subsectors becomes clear.

While financial institutions achieve ESG scores averaging 67.3 compared to non-financials' 54.2, this 24% advantage inverts when financed emissions enter calculations. Investment banks demonstrate the most extreme divergence, with FEPI scores of -0.89 indicating near-complete opacity regarding climate impact. This finding aligns with PCAF's documentation of 700x multipliers but extends understanding by quantifying how this blind spot translates into scoring advantages. This emission reality gap is summarized in Table 26 below.

Sector	ESG Score	Operational Emissions (tCO₂e)	Financed Emissions (tCO₂e)	True Multiplier	FEPI Score
Investment Banks	71.2	3,400	4,868,800	1,432x	-0.89
Commercial Banks	68.4	8,900	3,124,300	351x	-0.76
Asset Managers	69.8	1,200	892,400	744x	-0.82
Insurance	64.1	5,600	1,607,200	287x	-0.71
Manufacturing (comparison)	54.2	487,000	146,100	0.3x	0.43

Table 26 - The Emissions Reality Gap

Table 27 reveals 2015 Paris Agreement asymmetrical behavioral responses. The ensemble model identifies a structural break precisely at Q1 2016, with differential patterns that challenge conventional assumptions about regulatory effectiveness.

Metric	Financial Sector	Non-Financial Sector	Significance
Narrative Disclosure Volume	147%	67%	p<0.001
Quantitative Environmental Data	23%	89%	p<0.001
Scope 1-2 Completeness	12%	76%	p<0.001
Scope 3 Reporting	8%	41%	p<0.01
ESG Score Improvement	+18.3 points	+9.7 points	p<0.001
Actual Emissions Reduction	-2.10%	-18.40%	p<0.001

Table 27 - Paris Agreement Behavioral Shifts (2014-2016 vs 2017-2024)

This divergence exposes how Paris Agreement mechanisms inadvertently widened the measurement gap by rewarding disclosure practices over emission reductions. The improvements for financial firms, despite minimal emission reductions, validate the measurement-artifact hypothesis.

Clustering analysis reveals distinct ‘greenwashing clusters’ with troubling implications for market integrity. Financial firms concentrate in high-score yet low substance clusters, achieving composite ratings that systematically misrepresent environmental impact. Table 28 summarizes these ESG clustering patterns.

Cluster	(Financial/Non-Fin)	ESG Score	Governance	Environmental Data Completeness
1: "Transparent Polluters"	12/234	48.3	52.1	94%
2: "Improving Manufacturers"	8/389	56.7	61.4	78%
3: "Sophisticated Obscurers"	142/47	72.4	78.4	21%
4: "Governance Champions"	89/23	68.9	81.2	34%

Table 28 - ESG Clustering Patterns and Paradox Manifestation

Financial firms concentrate in Cluster 3 (n=142), characterized by high governance scores ($\mu=78.4$), moderate social scores ($\mu=61.2$), but critically absent environmental data (79% missing Scope 3). This cluster achieves composite ESG scores 31% higher than Cluster 1, where manufacturing firms with complete emissions data but display weaker governance structures.

The workforce composition analysis through 200,000+ ESG-related job postings reveals implementation gaps between stated commitments and organizational reality. Financial institutions allocate 73% of sustainability hiring to compliance and reporting roles versus 27% to operational transformation. Non-financial firms demonstrate inverse patterns (45% operational, 55% governance), suggesting substantive efforts despite a lower ESG score.

The synthetic scenario generator’s 10,000 Monte Carlo simulations quantify precise intervention thresholds required for paradox resolution. Table 29 below demonstrates that current incremental approaches fail, with linear interventions actually intensifying measurement distortions.

Intervention Type	<\$50/tCO₂	\$50-100/tCO₂	\$100-200/tCO₂	>\$200/tCO₂
Carbon Pricing Alone	0%	0%	8%	21%
+ Mandatory Scope 3	0%	12%	34%	67%
+ Methodology Reform	3%	28%	51%	89%
+ Enforcement (>£10M)	11%	43%	78%	96%

Table 29 - Policy Intervention Effectiveness

Carbon prices below \$100/tCO₂ widen the measurement gap by 18%, since penalties fall mainly on manufacturers' direct emissions while financial firms' financed emissions remain excluded. The paradox is only resolved in over half of simulated cases under drastic conditions. Carbon pricing above \$200/tCO₂ combined with mandatory Scope 3 disclosure and revised scoring methods that weight actual emissions at least 40%. This indicates that incremental policy tweaks are inadequate relative to the scale of distortion

Temporal dynamics reveal an acceleration of the paradox post 2020. The opacity index for financial firms increased from 0.52 to 0.71 between 2020 and 2024, while non-financial opacity decreased from 0.48 to 0.41. This divergence coincides with ESG disclosure regulations that paradoxically enable firms to leverage reporting infrastructure for scoring advantages without environmental improvements. The greenwashing detection algorithm flags 67% of financial institution sustainability reports as high risk for aspirational performance gaps, compared to 41% for non-financial firms, even after controlling for disclosure volume and linguistic complexity.

5. Discussion & Future Works

Theoretical Contributions

This research establishes the ‘ESG Opacity Paradox’ as a theoretical framework explaining how information asymmetry and methodological biases create systematic advantages for financial intermediaries. The paradox operates through three mutually reinforcing mechanisms:

- **Exclusion Blindness:** Missing financed emissions data paradoxically improves rather than penalizes scores, creating incentives for strategic non-disclosure.
- **Disclosure Dividend:** Disclosure quality proxies substitute for environmental performance, rewarding narrative complexity over substantive action.
- **Regulatory Capture:** Financial firms influence measurement standards that systematically favor service-sector business models over industrial operations.

This framework extends stakeholder theory by demonstrating how apparent alignment with stakeholder interest masks fundamental misalignment when measurement systems fail to capture material impacts. The dimensional instability revealed where ESG components show negative correlations for financial firms but positive correlations for non-financials, invalidating universal ESG frameworks and necessitating sector-specific assessment architectures.

Policy implications for the Bank of England and Regulatory Architecture

The findings carry direct implications for financial oversight and stability regulation. Evidence shows that current supervisory frameworks materially underestimate climate risk by rewarding disclosure practices rather than substantive impact reductions. Three regulatory interventions are suggested as critical.

First, mandatory Scope 3 reporting must extend beyond TCFD recommendations to include verified and audited financed emissions using standardized PCAF methodologies. Our analysis shows that voluntary regimes preserve the paradox by firms disclosing selectively while omitting the most material categories. Firms only disclose fully when rules are compulsory and the penalties are serious.

Second, a sectoral Impact-Weighted ESG framework can correct systematic biases in current scoring methodologies. Financial firms should be assessed with 60% weight on financed emissions, 25% on governance and 15% on operational metrics. Manufacturing firms require the inverse emphasis: 60% on operational emissions, 25% on supply chains,

and 15% on governance. Back testing shows this reweighting eliminates approximately 78% of paradoxical scoring advantages while maintaining high validity ($\alpha = 0.92$).

Third, integration of alternative data sources can overcome the inherent limitations of voluntary disclosure. Labor market signals, particularly the balance between compliance versus operational ESG hiring, act as forward-looking indicators, providing 6-18 months of lead time relative to reported ESG metrics. Workforce composition thus becomes a practical diagnostic tool for regulators to distinguish genuine transformation from disclosure-driven performance.

Beyond these interventions, the scenario generator provided three critical insights for policy design. Incremental carbon pricing below \$100/tCO₂ worsens the paradox: mandatory Scope 3 disclosure reduces but doesn't remove financial firms' advantage, and the paradox only gets resolved when strong carbon pricing, full transparency on emissions and fair rules are tailored to each industry.

Together, these insights underline that incremental disclosure or pricing adjustments are inadequate. To ensure financial system resilience, the Bank of England must adopt a comprehensive approach that integrates financed emissions, corrects sectoral weighting biases and incorporates forward-looking labor market signals into supervisory stress testing.

Methodological Innovations and Future Research Directions

This dissertation advances methodological frontiers in ESG research by integrating machine learning, natural language processing and labor market analytics within a unified framework. The architecture combines XGBoost for non-linear feature extraction, causal inference models (Difference-in-Differences, anomaly detection, missingness-as-signal), LSTM networks for temporal shifts and transformer-based NLP for textual disclosure analysis. Collectively, these methods allow for the deconstruction of ESG ratings into substance versus artifact, revealing that score variance is predominantly explained by opacity and disclosure patterns (74%) rather than emissions (2-3%). The resulting 0.68 R^2 in ESG prediction from largely artifactual features provides not only empirical validation of the 'measurement paradox' but also a methodological template for future research that treats missingness as signal rather than nuisance.

A second innovation lies in the Synthetic ESG Scenario Generator, which employs Monte Carlo simulations to stress-test policy interventions. Unlike static retrospective analyses, this engine introduces counterfactual policy shocks (carbon pricing, mandatory Scope 3 disclosure, greenwashing crackdowns) and quantifies their impact on sectoral ESG

trajectories. By simulating over 10,000 firm-years, the framework produces P10/P50/P90 distributions that highlight thresholds where incremental policies fail, and only systemic interventions, carbon prices above \$200/tCO₂ coupled with disclosure mandates and sectoral reweighting, meaningfully resolve the paradox. This scenario-based sandbox introduces a reproducibility standard absent in prior ESG stress testing and offers policymakers a forward-looking diagnostic tool.

Third, the research introduces labor market signals as leading indicators of sustainability performance. By processing 224,000+ ESG-related job postings, the study demonstrates that workforce composition predicts ESG trajectories with a 6-18 month lead time. Non-financial firms' hiring in climate and operational functions correlates with substantive emissions reduction (+0.59 ESG points per 10% increase, $p < 0.001$), whereas financials' governance-heavy hiring yields marginal returns, reflecting disclosure saturation. This methodological strand addresses a structural weakness in ESG research. The lag between reported commitments and observable implementation. By embedding human capital dynamics into ESG assessment, this study contributes a novel dimension that links talent allocation to measurable outcomes.

Despite these advances, boundary conditions remain. The analysis is constrained by **80% Scope 3 missingness** in financial firms, which necessitated imputation. Sensitivity tests confirm robustness within $\pm 30\%$ variation, but extreme assumptions could alter paradox magnitude. Similarly, while the study covers 2010–2024 and captures the Paris Agreement as a structural break, other inflection points such as the 2008 financial crisis and 2020 COVID-19 pandemic suggest that ESG measurement dynamics are crisis-sensitive. Extended historical back casting to 2000 or inclusion of post-2025 transition finance reforms would enhance temporal generalization.

Private markets constitute another blind spot. With \$4.7 trillion in private equity assets under management, these firms face distinct disclosure incentives. Preliminary evidence suggests a 12% deterioration in ESG scores post-delisting, implying that public-market opacity may understate private-market distortions. Incorporating private equity and venture capital data would be a critical extension of this research.

Future research can extend this framework across three dimensions:

1. **Geographic Replication:** Cross-jurisdictional studies are essential to test whether the paradox is an Anglo-American pathology or a universal measurement flaw. Preliminary EU results suggest attenuation under SFDR (EU Sustainable Finance Disclosure Regulation), but paradox persistence ($p < 0.05$) indicates that regulation mitigates but does not eliminate bias. Expansion to Asia-Pacific and emerging

markets would clarify whether regulatory density or institutional maturity drives these differences.

2. **Temporal Acceleration:** Real-time ESG assessment using high-frequency alternative data such as **satellite imagery, supply-chain IoT, and sentiment streams** could potentially address the lag that enables greenwashing. Pilot integration in this study shows feasibility, but scaling requires validation of reliability and interpretability. Coupling such feeds with **blockchain-based disclosure registries** could ensure immutability, while smart contracts could automate the enforcement of aspiration-performance gaps.
3. **Methodological Deepening:** The adoption of **causal machine learning** (double/debiased ML, causal forests) promises sharper identification of genuine sustainability drivers. Early causal forest results suggest heterogeneous Paris Agreement effects, with smaller firms and ESG laggards showing substantive gains, while large financials optimized disclosure. Scaling such methods could reveal whether policy effectiveness varies systematically by firm size, geography, or ownership structure.

Overall, the innovations presented here reinvent ESG assessment by reframing opacity, missingness, and greenwashing not as obstacles but as measurable artifacts of the system. Future research must push further into **cross-market validation, crisis-resilient frameworks, and implementation diagnostics** that move beyond ratings toward outcome-linked accountability. Without such advances, ESG will remain a scoreboard that rewards narrative fluency over environmental impact.

6. Conclusion

The study finds that financial firms' higher ESG ratings don't reflect true sustainability, but rather a systematic advantage created by the way ESG performance is measured.

Revolving the Central Question

The empirical evidence converges on an uncomfortable truth. Financial institutions achieve ESG scores averaging 67.3 compared to non-financials' 54.2, yet this 24% advantage disappears when financed emissions enter calculations. Investment banks, the supposed sustainability leaders with ESG scores of 71.2, carry financed emissions 1,432 times their operational footprint, yielding Financed Emissions Performance Index scores of -0.89. This is not measurement error; it is measurement design operating precisely as constructed.

The ESG Opacity Paradox framework developed herein reveals three mutually reinforcing mechanisms that sustain this illusion. Exclusion blindness rewards the absence of Scope 3 data with 6.2 additional ESG points. Disclosure dividends substitute narrative complexity for environmental substance, with financial firms producing 147% more disclosure volume while providing 66% less quantitative environmental data. Regulatory capture enables financial institutions to shape measurement standards that weight governance at 40% while emissions receive 8% in sector-specific frameworks.

The Paris Agreement amplified rather than resolved the paradox, with financials rewarded for disclosure while non-financials delivered real emission cuts. Non-financials cut emissions by 18.4% but with only a 9.7 point score gain, revealing that existing frameworks favor regulatory artifacts rather than genuine sustainability practices.

Labor market signals confirm this implementation gap. Financial institutions allocate 73% of ESG hiring to compliance and reporting roles, while non-financials direct 45% toward operational transformation. The causal inference framework quantifies that the consequence of climate-focused hiring generates 0.59 ESG points per 10% increase in non-financial firms but yields negligible improvements in financial firms, indicating saturation of disclosure-based scoring mechanisms.

The Path Forward

The current evidence leaves no ambiguity. Continuing current trajectories will establish rather than resolve the measurement paradox. Financial markets are allocating trillions based on signals that systematically misrepresent climate risk, intensifying systemic vulnerabilities with every delay. For the Bank of England, the findings reveal a crucial blind spot. Current stress tests exclude financed emissions and, therefore, true climate risks are underestimated. The framework developed in this study offers regulators practical

instruments such as the opacity index, greenwashing detection score, a predictive dashboard with ESG, disclosure documents and labor market analysis, which separates substance from artifacts. Until financed emissions are fully integrated and sectoral biases are corrected, ESG ratings will continue inverting reality, directing capital to those who finance climate risk while penalizing those bearing transition costs. The opacity paradox is not an accidental error but deliberate design. Resolving it demands replacing today's disclosure-centric architecture with measures that capture outcomes rather than appearances.

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8. Appendix

1. Regulatory events utilized in the Regulatory Events Framework

Policy / Framework	Year	Region	Sector Target
Paris Agreement	2015	Global	All
TCFD Mandatory Disclosure	2021	UK	Financial
CSRD	2021	EU	Non-Financial
SDR (UK FCA)	2024	UK	Financial
SEC Climate Disclosure (Draft/Proposal)	2022	US	All
GRI Standards (G3.1, Universal)	2011 / 2021	Global	Non-Financial
EU Taxonomy / NFRD / CSRD	2014–2022	EU	All / Non-Financial
UK Green Finance Strategy	2023	UK	All
UK Companies Act (Env. Reporting)	2010	UK	All
CDP Scope 3 Reporting	2013	Global	Financial
BoE Climate Risk Expectations	2019	UK	Financial
SASB Standards (now ISSB)	2018	US	All
IFRS/ISSB Standards (S1 & S2)	2023	Global	All
Regional ESG Rules (Japan GPIF, HKEX, MAS, Asia-Pacific)	2016–2021	Asia	Financial / All
UN PRI Mainstreaming	2010	Global	Financial

2. SQL workflow used to identify ESG-related job postings, remove false positives, deduplicate multi-source records, and build firm/region/sector/temporal aggregates.

2.1 Source tables (BigQuery) and key deliverables

Table	Purpose
jobs_filtered_v1	Title and description based ESG keyword filter.
jobs_deduplicated	Cross-source deduplication via composite ID / fingerprint.
job_esg_filtered_clean_v2	Merge filters, remove false positives, score relevance, high-confidence flag.
company_esg_metrics	Company-level counts, % ESG jobs, focus categories.
regional_esg_metrics	Region/continent adoption, focus mix, averages.
country_esg_metrics	Country adoption, mix, and intensity.
sector_esg_comparison	Sector adoption rate, intensity, focus shares.
temporal_esg_metrics	Quarterly trend, QoQ growth by sector.
esg_adoption_summary	Summaries for comparative analysis & top companies.

2.2 ESG Filtering Keywords (159 terms across 7 categories)

Category	# Terms	Examples (Representative)
Core ESG	6	ESG, sustainability, CSR, sustainable development
Climate	15	climate change, net zero, carbon footprint, Paris Agreement
Environmental	14	renewable, biodiversity, recycling, water management
Regulatory	14	TCFD, CSRD, SFDR, GRI, CDP, EU Taxonomy, ISSB, TNFD
Social	9	DEI, human rights, supply chain sustainability, SRI
Finance/Energy	9	green finance, sustainable finance, energy transition
Title-specific	23	ESG Analyst, Head of Sustainability, Net Zero Officer

2.3 False Positive Keywords (19 exclusions)

Category	Examples
IT / Software	development environment, production environment, test environment, green screen
Administrative	carbon copy, carbon paper, renewable contract, renewable term
Immigration	green card
Marketing / Sales	social media, social network, social selling

2.4 Additional Classification Keywords

Category	# Terms	Examples
Financial sector IDs	10	bank, insurance, asset management, fintech, hedge fund
Non-financial sector IDs	10	manufacturing, energy, retail, healthcare, technology
Seniority keywords	10	Chief Sustainability Officer, VP Sustainability, Analyst
Geographic regions	7 sets	Europe, North America, Asia-Pacific, Latin America, Africa, Middle East, Oceania (60+ countries)

2.5 Scoring Methodology

Component	Weight	Rule
Core ESG keywords	2x	Double weight as fundamental indicators
Climate / Env / Social / Finance / Regulatory	1x	Standard weight each
Total divisor	7	Normalized score calculation
High-confidence flag	—	Triggered when <i>core</i> OR <i>regulatory</i> keywords are present

- All mathematical formulas, statistical models, and computational methods used in this research.

3.1 Carbon Intensity Metrics

Name	Formula	Unit
Carbon Intensity per Revenue	$CI_{Rev} = CO2_{TotalRevenue} / 10^6 CI_{\{Rev\}}$ $= \frac{CO2_{\{Total\}}}{\{Revenue / 10^6\}}$	tCO ₂ / M\$
Carbon Intensity per Assets	$CI_{Assets} = CO2_{TotalTotalAssets} / 10^6 CI_{\{Assets\}}$ $= \frac{CO2_{\{Total\}}}{\{TotalAssets / 10^6\}}$	tCO ₂ / M\$
Carbon Intensity per Market Cap	$CI_{MCap} = CO2_{TotalMarketCap} / 10^6 CI_{\{MCap\}}$ $= \frac{CO2_{\{Total\}}}{\{MarketCap / 10^6\}}$	tCO ₂ / M\$

3.2 Composite Indices

Name	Formula	Description
FEPI – Financed Emissions Performance Index	$FEPI = ROA \times ESG100 \times \ln(CO2 + e)$ $FEPI = ROA \times \frac{ESG}{100} \times \frac{1}{\ln(CO2 + e)}$	Carbon-adjusted performance; key paradox metric
Sustainability Alpha	$\alpha_{sus} = 0.5(ESG - \mu_{ESG}) - 0.5(CI - \mu_{CI}) \alpha_{sus}$ $= 0.5(ESG - \mu_{ESG}) - 0.5(CI - \mu_{CI})$	Sector-relative outperformance
Opacity Index	$OI = 0.35M + 0.25F + 0.20N + 0.20TOI$ $= 0.35M + 0.25F + 0.20N + 0.20T$	Composite of missing emissions (M), financial gaps (F), narrative gaps (N), temporal inconsistency (T)
Greenwashing Score	$GW = 0.25OI + 0.20MRG + 0.20SID$ $+ 0.15SDI + 0.10CP$ $+ 0.10PPGGW$ $= 0.25OI + 0.20MRG + 0.20SID + 0.15SDI$ $+ 0.10CP + 0.10PPG$	Composite deception metric

3.3 Regulatory Pressure Metrics

Name	Formula	Unit
Regulatory Pressure Index	$RPI(t) = \Sigma (Weight \times Proximity \times Enforcement) RPI(t)$ $= \Sigma (Weight \times Proximity \times Enforcement)$	Index

4. Code Repository: The POLARIS dashboard, hosted via Streamlit, is available at the following GitHub repository.

Link: <https://github.com/Shanila3d/POLARIS.git>